**PHINMA UNIVERSITY OF ILOILO**

College of Information Technology Education

# CAPSTONE PROJECT PROPOSAL

## FlexSpace: Your Gateway to Co-Working Spaces in Iloilo City

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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CHAPTER 1

# INTRODUCTION

## 1.1 Background of the Study (Expanded)

The emergence of coworking spaces is not merely a passing trend but a structural transformation in how societies organize work. Globally, coworking hubs have evolved from small, community-driven initiatives into sophisticated ecosystems that support entrepreneurship, innovation, and academic productivity. In cities such as Singapore, Berlin, and San Francisco, coworking has become synonymous with start-up culture, providing flexible infrastructure for digital nomads, freelancers, and small enterprises. These hubs are often integrated into broader Smart City frameworks, where digital technologies, urban planning, and socio-economic policies converge to create sustainable and inclusive work environments.

Growth of Coworking Spaces in Iloilo City

In the Philippine context, coworking spaces have proliferated in metropolitan centers such as Metro Manila and Cebu, driven by the rise of the gig economy and the increasing demand for flexible work arrangements. Iloilo City, recognized as one of the country's emerging digital hubs, has begun to follow this trajectory. The city's Smart City initiatives, which emphasize digital transformation, sustainable urban development, and inclusive growth, provide fertile ground for coworking ecosystems. Districts such as Mandurriao, with its burgeoning business parks, and Jaro, with its academic institutions, represent diverse socio-spatial contexts where coworking

spaces can thrive. The expansion of coworking hubs in Iloilo reflects not only economic growth but also cultural shifts toward collaborative and flexible work practices.

## Rise of Freelancers in the Philippines

The rise of freelancing in the Philippines has further accelerated the demand for coworking spaces. Online platforms such as Upwork, Fiverr, and OnlineJobs.ph have enabled Filipino professionals to secure international contracts, creating a workforce that is highly mobile and digitally connected. Freelancers often require reliable internet, professional environments, and flexible schedules—needs that traditional offices or home setups cannot consistently meet. In Iloilo City, freelancers represent a growing demographic that views coworking hubs as essential infrastructure for productivity and competitiveness in the global digital economy.

## Students Needing Study Spaces

Students constitute another critical demographic driving coworking demand. In districts such as Jaro, which hosts several academic institutions including PHINMA University of Iloilo, the need for study-friendly environments is particularly acute during examination seasons. Libraries and campus facilities often reach capacity, leaving students without conducive spaces for learning. Coworking hubs provide affordable access to quiet, well-equipped environments that support academic productivity. By offering high-speed internet, collaborative rooms, and accessible locations, these hubs bridge the gap between institutional resources and student needs.

## Digital Transformation of Local Businesses

Local businesses in Iloilo are also undergoing digital transformation, adopting e-commerce platforms, cloud-based tools, and online marketing strategies. However, many small and medium enterprises (SMEs) lack the physical infrastructure to support these digital initiatives. Coworking spaces provide SMEs with access to professional facilities without the financial burden of long-term leases. They also foster networking opportunities, enabling entrepreneurs to collaborate, share resources, and innovate. In this way, coworking hubs contribute to the broader economic resilience and modernization of Iloilo's business sector.

## Why Reservation Systems Matter

Despite the growth of coworking spaces, the infrastructure for discovering and reserving them in Iloilo remains fragmented. Most hubs rely on manual systems such as Facebook Messenger inquiries, phone calls, or physical walk-ins. These methods are inefficient and prone to errors, leading to double bookings, wasted time, and user dissatisfaction. For students balancing academic schedules, freelancers managing client deadlines, and entrepreneurs coordinating team meetings, such inefficiencies can have significant consequences. Globally, reservation systems have evolved to integrate IoT sensors, machine learning algorithms, and GIS mapping, providing real-time occupancy data and transparent pricing. Iloilo's reliance on manual updates underscores the need for a localized solution that aligns with global best practices while addressing the city's socio-economic realities.

## Technological and Policy Context

Technologically, the MERN stack offers a robust foundation for building such a system. MongoDB provides flexible data storage for hub listings and reservations, Express.js and Node.js enable scalable server-side operations, and React ensures a responsive, user-friendly interface. When combined with GIS mapping, the system can visualize coworking hubs across Iloilo's districts, allowing users to filter by location, amenities, and availability. For hub owners, administrative dashboards powered by CRUD operations and middleware authentication provide automated tools for managing occupancy and optimizing resource allocation. From a policy perspective, the system supports Iloilo's Smart City agenda by digitizing urban resource management. By reducing inefficiencies in coworking reservations, the platform contributes to broader goals of sustainability, inclusivity, and innovation. It also empowers local hub owners with data-driven insights, enabling them to make informed decisions about facility usage, pricing strategies, and service improvements.

## Summary

In summary, the background of this study situates coworking spaces within global, national, and local contexts, highlighting both the opportunities and challenges facing Iloilo City. It establishes the rationale for developing a Co-Working Space Discovery and Reservation System that leverages modern technologies to address persistent inefficiencies, thereby supporting the city's transformation into a Smart City. By addressing the needs of freelancers, students, SMEs, and local government, the proposed system represents not only a technological innovation but also a

socio-economic intervention that enhances productivity, inclusivity, and urban resilience.

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## 1.2 Statement of the Problem

The central difficulty addressed by this study lies in the persistent reservation uncertainty and information fragmentation experienced by remote workers, freelancers, and students in Iloilo City. While coworking spaces have proliferated in recent years, their operational models remain largely dependent on manual, non-centralized inquiry methods. These include social media messaging, phone calls, and physical walk-ins, which, although familiar, are inefficient and prone to error. The following subsections elaborate on the specific problems, their causes, and their impacts on users, owners, and the broader Smart City agenda.

### Reservation Uncertainty

Reservation uncertainty manifests when users attempt to secure a workspace without reliable information about current occupancy. In Iloilo City, anecdotal evidence from students and freelancers reveals that arriving at full-capacity venues is a common occurrence. This phenomenon aligns with Kumar and Patel's (2023) findings that manual booking systems frequently lead to double reservations and user frustration. The root cause of this uncertainty lies in the absence of real-time availability tracking. Hub owners often rely on paper logs or spreadsheets, which are updated irregularly and fail to reflect actual occupancy. For knowledge workers whose productivity

depends on uninterrupted access to conducive environments, such uncertainty translates into wasted time, missed deadlines, and diminished trust in coworking providers. For example, a freelancer in Mandurriao may travel across the city only to discover that all seats are occupied, resulting in lost billable hours and client dissatisfaction.

**Information Fragmentation**

Information fragmentation further compounds the problem. Pricing, amenities, and availability data are scattered across disparate channels, often limited to Facebook posts or informal word-of-mouth networks. This lack of transparency prevents users from making informed decisions about which coworking hub best suits their needs. In districts such as Mandurriao, where multiple hubs compete for clientele, the absence of a unified platform creates inefficiencies not only for users but also for owners, who struggle to differentiate their services in a crowded market. Students in Jaro, for instance, may be unaware of affordable study hubs nearby because information is not centralized, leading them to overcrowded libraries or less conducive environments.

**Booking Conflicts and Operational Inefficiencies**

Manual tracking systems also lead to booking conflicts, where multiple users reserve the same seat or room due to outdated information. Hub owners, relying on spreadsheets or paper logs, face difficulties in managing occupancy, resulting in scheduling overlaps and dissatisfied clients. These inefficiencies undermine the credibility of coworking spaces as reliable infrastructures for professional and

academic productivity. For owners, the consequences include lost revenue opportunities, reputational damage, and reduced competitiveness. For users, the impact is equally severe, as booking conflicts disrupt workflow and erode confidence in the reliability of coworking hubs.

**Lack of Unified Platform**

The absence of a centralized discovery and reservation system represents a structural gap in Iloilo's coworking industry. Unlike global cities that have adopted digital platforms integrating IoT sensors, GIS mapping, and machine learning algorithms, Iloilo continues to rely on fragmented, manual processes. This gap not only limits user choice but also constrains the city's ability to position itself as a competitive digital hub within the Philippines and Southeast Asia. Without a unified platform, Iloilo risks lagging behind in its Smart City trajectory, undermining its potential as a regional startup hub.

**Theoretical Framing**

From a socio-technical perspective, coworking spaces are ecosystems where technological infrastructures and social practices intersect (Amler, 2024). The inefficiencies observed in Iloilo reflect a misalignment between the city's Smart City ambitions and the operational realities of its coworking industry. Theories of urban innovation (Kraus et al., 2022) suggest that digital platforms are essential for embedding coworking into the broader urban fabric, enabling equitable access and

sustainable growth. Without such platforms, Iloilo risks perpetuating inequities in resource distribution and limiting its capacity for inclusive digital transformation.

**Local Case Examples**

In Mandurriao, known for its burgeoning business parks, freelancers often rely on Facebook Messenger to inquire about seat availability. Reports indicate that responses are delayed, leading to wasted travel time and missed opportunities. In Jaro, where academic institutions drive demand for study hubs, students frequently encounter booking conflicts due to manual reservation logs. In Molo and the City Proper, smaller coworking spaces struggle to attract clients because their services are not visible on centralized directories. These local examples illustrate the pervasive nature of reservation uncertainty and information fragmentation across Iloilo's districts.

**Implications**

The implications of these problems are multifaceted. For users, inefficiencies translate into lost productivity, financial costs, and diminished trust in coworking providers. For hub owners, operational inefficiencies hinder revenue optimization and customer satisfaction. For the city, the absence of a centralized system undermines its Smart City agenda, limiting its ability to leverage coworking as a driver of digital innovation and economic growth. Addressing these problems is therefore not only a technological imperative but also a socio-economic necessity for Iloilo City's sustainable development.

## 1.3 Objectives of the Study

### 1.3.1 General Objective

The general objective of this study is to design and develop a **centralized Co-Working Space Discovery and Reservation System for Iloilo City** that streamlines the search for available workspaces, provides real-time seat tracking, and facilitates secure bookings for professionals and students. This overarching goal reflects both a technological ambition and a socio-economic imperative: to digitize the fragmented coworking ecosystem of Iloilo City and align it with global best practices in Smart City innovation. By creating a unified platform, the study seeks to eliminate inefficiencies in manual reservation processes, enhance user trust, and empower hub owners with data-driven management tools.

### 1.3.2 Specific Objectives

1. To achieve this general objective, the study pursues several specific objectives, each of which contributes to the system's overall functionality and relevance.

2. **1. Analyze the current challenges faced by the Iloilo workforce regarding workspace accessibility and the limitations of manual reservation processes.**

3. This objective ensures that the system design is grounded in empirical realities. By examining the experiences of students, freelancers, and entrepreneurs, the study identifies pain points such as reservation uncertainty, booking conflicts, and information fragmentation. This analysis provides the foundation for designing solutions that are responsive to local needs rather than abstract technological ideals.

4. **2. Design an intuitive and responsive user interface that allows users to filter spaces by district, amenities, and pricing.**

5. User interface (UI) and user experience (UX) design are critical determinants of system adoption. A responsive interface ensures accessibility across devices, particularly smartphones, which are the primary digital tools for many Ilonggos. By enabling filters for districts (Mandurriao, Jaro, Molo, City Proper), amenities (e.g., high-speed Wi-Fi, meeting rooms), and pricing, the system empowers users to make informed decisions quickly and efficiently. This objective aligns with ISO 25010's emphasis on usability as a dimension of software quality.

6. **3. Develop a real-time availability tracking module to monitor desk and room occupancy, preventing overbooking and walk-in rejections.**

7. Reservation uncertainty is one of the most pressing challenges in coworking ecosystems. By implementing a real-time tracking module, the system ensures that users have accurate information about seat and room

availability. Although the study acknowledges its limitation in not deploying IoT hardware, software-based status updates and manual admin overrides provide a feasible alternative. This objective directly addresses the problem of overbooking and enhances user trust in the system.

8. **4. Implement an automated notification and reservation system that sends instant booking confirmations and reminders to users.**

9. Automation reduces human error and enhances efficiency. By sending instant confirmations and reminders, the system minimizes booking conflicts and ensures that users are informed about their reservations. This feature reflects the principles of agile system design, where responsiveness and reliability are prioritized.

10. **5. Create an administrative dashboard for coworking space owners to manage their inventory, update facility information, and view usage analytics.**

11. Hub owners are critical stakeholders in the coworking ecosystem. Administrative dashboards provide them with tools to manage occupancy, update listings, and analyze usage patterns. Through CRUD operations and middleware authentication, owners can maintain data integrity while optimizing facility usage. This objective empowers owners with actionable insights, enabling them to adjust pricing strategies, identify peak usage times, and improve customer satisfaction.

12. **6. Integrate a mapping API to provide accurate geolocations and navigation for various coworking hubs across Iloilo City.**

13. Geographic accessibility is a key factor influencing coworking adoption. By integrating a mapping API, the system visualizes coworking hubs across Iloilo's districts, enabling users to identify nearby facilities and navigate to them efficiently. This objective aligns with Smart City principles of transparency and inclusivity, embedding spatial data into citizen-centric services.

14. **7. Store and manage user profiles, historical booking data, and facility details using a secure database management system.**

15. Data management is essential for both functionality and security. By storing user profiles, booking histories, and facility details in MongoDB, the system ensures flexibility and scalability. Middleware authentication safeguards data integrity, while secure database practices protect user privacy. This objective reflects ISO 25010's emphasis on reliability and security.

16. **8. Test and evaluate the system's reliability, responsiveness, and usability based on standard software quality metrics.**

17. Evaluation is critical for academic rigor. By testing the system against ISO 25010 standards, the study ensures that its findings are credible and replicable. Usability tests, surveys, and data analysis provide empirical

evidence of the system's effectiveness, demonstrating its contribution to both academic discourse and practical innovation.

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### 1.4 Scope and Delimitation

Every research project must clearly define its scope to establish boundaries of inquiry and ensure feasibility. For this study, the scope encompasses the design, development, and evaluation of a **localized web and mobile-based application** for discovering and reserving coworking spaces specifically within Iloilo City. By narrowing the geographic and functional focus, the study ensures that its objectives are achievable within the academic timeframe and resources available, while still contributing meaningfully to both scholarly discourse and practical innovation.

#### Scope of the Study

The system will include several core modules:

- **User Registration and Profiles.** Users such as students, freelancers, and entrepreneurs will be able to create accounts, store booking histories, and manage preferences. This feature ensures personalization and continuity of service.
- **Searchable Directory of Hubs.** The system will provide a centralized directory of coworking spaces across Iloilo's districts (Mandurriao, Jaro, Molo,

City Proper). Each listing will include amenities, pricing, and availability, enabling users to make informed decisions.

- **Real-Time Seat and Room Availability Tracking.** Through software-based status updates and manual admin overrides, the system will provide users with accurate information about occupancy, reducing reservation uncertainty and walk-in rejections.
- **Reservation Engine.** The backend logic will process booking requests, check for scheduling conflicts, and confirm time slots. Automated notifications will send instant confirmations and reminders, minimizing human error.
- **Administrative Dashboard.** Hub owners will have access to dashboards that allow them to manage inventory, update facility information, and view usage analytics. This feature empowers owners with data-driven insights, enabling them to optimize operations and revenue.
- **Mapping API Integration.** Geographic Information System (GIS) or mapping APIs will visualize coworking hubs across Iloilo City, providing accurate geolocations and navigation. This feature enhances accessibility and supports Smart City principles of transparency and inclusivity.

Together, these modules define the functional scope of the system, ensuring that it addresses both user and owner needs while aligning with Iloilo City's Smart City agenda.

**Delimitations of the Study**

While the scope is ambitious, the study also acknowledges several delimitations that constrain its functionality:

1. **Exclusion of IoT Hardware.** The system will not include real-time IoT-based hardware integration, such as pressure sensors or RFID systems for seat detection. Instead, it will rely on software-based status updates and manual admin overrides. This delimitation reflects resource constraints but also ensures feasibility within an academic context.
2. **Simulated Payment Gateway.** The payment gateway will be implemented as a simulated environment for academic purposes. This means that the system cannot process actual financial transactions, limiting its applicability in commercial deployment. However, the simulation provides a foundation for future integration of secure payment systems.
3. **Geographic Limitation.** The database will be restricted to registered coworking spaces and study cafes within Iloilo City limits. Peripheral municipalities and long-term corporate lease management systems are excluded. This delimitation ensures focus on the most relevant local context.
4. **Exclusion of Corporate Lease Management.** The system will not include modules for managing long-term office rentals or corporate leases. Its focus is on short-term reservations for freelancers, students, and entrepreneurs.

## 1.5 Significance of the Study

The proposed Co-Working Space Discovery and Reservation System for Iloilo City holds significance across multiple stakeholder groups. Its impact extends beyond individual users to coworking hub owners, the local government, academic institutions, tourists, and future researchers. By digitizing the fragmented coworking ecosystem, the system contributes to both practical innovation and scholarly discourse, aligning with Iloilo City's Smart City agenda.

### Freelancers and Remote Workers

For freelancers and remote workers, coworking spaces represent more than physical infrastructure; they are enablers of productivity, creativity, and professional independence. Yet, reservation uncertainty and information fragmentation often undermine their ability to secure reliable workspaces. The proposed system provides a "one-stop shop" for discovering and reserving coworking hubs, reducing the frustration of walk-in rejections and wasted travel time. By offering real-time occupancy data, transparent pricing, and amenity filters, the system empowers freelancers to make informed decisions quickly. This significance is amplified in Iloilo City, where the gig economy is expanding and digital nomadism is becoming more prevalent.

### Students

Students represent a critical demographic in Iloilo City, particularly in districts such as Jaro, which hosts several academic institutions. During exam seasons, demand for

study-friendly hubs increases, yet manual reservation systems often fail to meet this demand. The proposed system addresses this gap by identifying affordable and quiet coworking spaces with available seating. By integrating GIS mapping, the system ensures that students can locate nearby hubs efficiently, reducing travel time and enhancing academic productivity. In this way, the system contributes to educational outcomes by providing students with reliable access to conducive learning environments.

**Co-Working Space Owners**

Hub owners are essential stakeholders in the coworking ecosystem. Operational inefficiencies such as double bookings, underutilized resources, and limited visibility hinder their ability to optimize revenue and customer satisfaction. The administrative dashboards provided by the system offer automated tools for managing inventory, updating facility information, and analyzing usage patterns. Through CRUD operations and middleware authentication, owners can maintain data integrity while generating actionable insights. For example, usage analytics can reveal peak demand periods, enabling owners to adjust pricing strategies or expand capacity. By increasing visibility to a wider Ilonggo market, the system enhances competitiveness and sustainability for coworking businesses.

**Local Government of Iloilo City**

The local government stands to benefit from the system as part of its broader Smart City initiatives. By digitizing coworking reservations, the platform contributes to urban

resource optimization, reducing inefficiencies and fostering innovation. Transparent data on coworking availability supports evidence-based policymaking, enabling the government to identify underserved districts and allocate resources accordingly. Moreover, the system strengthens Iloilo's position as a digital hub, attracting entrepreneurs, start-ups, and investors. In this way, the study contributes to economic development, sustainability, and inclusivity, aligning with the city's vision of becoming a Smart City.

**Tourists and Digital Nomads**

Tourists and digital nomads represent an emerging user group for coworking spaces in Iloilo City. Short-term visitors often require professional environments to conduct business, attend virtual meetings, or complete freelance projects. The proposed system enhances their experience by providing accessible information on available hubs, amenities, and pricing. By integrating geolocation features, the system ensures that visitors can easily identify nearby coworking spaces, thereby supporting Iloilo's tourism and hospitality industries. This significance extends to the city's branding as a destination for both leisure and professional engagement.

**BSIT Students and Future Developers**

For Bachelor of Science in Information Technology (BSIT) students, the study provides a practical application of full-stack web development, database management, and API integration. By working with the MERN stack, students gain hands-on experience in modern web technologies, preparing them for careers in

software development and systems design. The study also exposes students to Agile methodology and ISO 25010 software quality standards, ensuring that their learning is aligned with industry best practices. Future developers may build upon this work by integrating IoT hardware, secure payment gateways, or expanding the geographic scope beyond Iloilo City. In this way, the project serves as both an academic requirement and a professional training ground.

**Researchers and Scholars**

Finally, the study serves as a foundational reference for future research on localized resource management systems and Smart City solutions. By documenting the design, implementation, and evaluation of the system, the study provides a blueprint that can be replicated or expanded in other contexts. Future researchers may build upon this work by integrating advanced technologies such as artificial intelligence, blockchain-based reservation systems, or expanding the geographic scope to include other Philippine cities. In this way, the study contributes to the evolving discourse on coworking ecosystems, digital platforms, and urban innovation.

## 1.6 **THEORETICAL FRAMEWORK**

**Technology Acceptance Model (TAM)**

The Technology Acceptance Model (TAM), developed by Davis (1989), explains how users come to accept and use technology. The model emphasizes two primary factors: perceived usefulness and perceived ease of use. Perceived usefulness refers

to the extent to which a user believes that using a system will improve performance, while perceived ease of use refers to the extent to which the system is free from effort.

In the context of the Co-Working Space Discovery and Reservation System for Iloilo City, TAM is highly relevant because the success of the system depends on whether students, freelancers, entrepreneurs, and coworking space owners are willing to adopt the platform. If users perceive that the system saves time, reduces booking conflicts, and provides convenient reservation services, they are more likely to use it consistently.

The study integrates TAM by ensuring that the system interface is user-friendly, responsive, and accessible across devices. Features such as real-time availability tracking, instant booking confirmation, and GIS-based navigation contribute to perceived usefulness. Meanwhile, responsive design, simplified navigation, and intuitive booking workflows contribute to perceived ease of use.

**Smart City Theory**

Smart City Theory emphasizes the integration of digital technologies into urban management to improve efficiency, sustainability, and quality of life. Smart cities leverage information systems, data analytics, and digital infrastructures to optimize public services and urban resources.

Iloilo City has been recognized for its Smart City initiatives, particularly in digital transformation and sustainable urban development. The proposed coworking

reservation system contributes to this agenda by digitizing the management of coworking spaces and providing a centralized platform for resource optimization.

Through real-time reservation management, GIS mapping, and centralized information access, the proposed system aligns with Smart City principles by improving accessibility, reducing inefficiencies, and supporting innovation-driven urban development.

**Socio-Technical Systems Theory**

Socio-Technical Systems Theory states that organizational systems involve both social and technical components that must function together harmoniously. Technology alone cannot solve problems unless it aligns with human behaviors, organizational practices, and environmental contexts.

In the proposed study, coworking spaces represent socio-technical ecosystems where users, owners, technologies, and urban infrastructures interact. The reservation system does not only provide technological functionality but also facilitates communication, collaboration, and accessibility among users and coworking providers.

The theory guides the development of a platform that addresses both technical requirements and user experiences. It ensures that the system is designed not only for operational efficiency but also for usability, inclusivity, and collaboration.

**Information Systems Success Model**

The Information Systems Success Model by DeLone and McLean emphasizes six dimensions of successful systems: system quality, information quality, service quality, use, user satisfaction, and net benefits.

The proposed coworking reservation system adopts this framework to evaluate its effectiveness. System quality is measured through usability and reliability, information quality is assessed through accuracy and real-time updates, while user satisfaction is evaluated through surveys and ISO 25010 testing.

The framework supports the evaluation phase of the study by providing measurable indicators of system success.

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### 1.7 Definition of Terms

Clarity of terminology is essential in any scholarly work, particularly in studies that integrate technical concepts with socio-economic contexts. The following terms are defined not only in their general sense but also in relation to the Co-Working Space Discovery and Reservation System for Iloilo City, ensuring that readers understand their specific application within this study.

**Reservation**

- Reservation refers to the formal act of securing a desk, seat, or room in advance for a specific time slot. Within this study, reservations are processed through the system's backend logic, ensuring that users receive instant confirmations and minimizing the risk of double bookings. By digitizing this process, the system reduces inefficiencies associated with manual walk-ins and enhances user trust in coworking hubs. This feature is particularly relevant in Iloilo City, where students and freelancers often face uncertainty when arriving at full-capacity venues.

**Booking**

- Booking is the process of recording a reservation into the system's database, thereby creating a binding agreement between the user and the coworking hub. Unlike informal inquiries through social media or phone calls, bookings provide assurance of availability and allow hub owners to manage occupancy more effectively. In this study, the booking engine is designed to prevent scheduling conflicts and ensure transparency, aligning with Smart City principles of reliability and inclusivity.

**Walk-In**

- A walk-in describes a user who arrives at a coworking hub without prior reservation. While common in Iloilo City, walk-ins often lead to inefficiencies such as overcrowding, wasted travel time, or outright rejection due to full capacity. The proposed system aims to reduce reliance on walk-ins by promoting digital reservations, thereby improving user experience and optimizing resource allocation for hub owners.

**QRPH (Quick Response Payment Philippines)**

- QRPH is the standardized QR code payment system adopted in the Philippines to facilitate secure, cashless transactions. Although the project uses a simulated payment gateway for academic purposes, QRPH integration represents a future enhancement that would allow users to pay for coworking services seamlessly. Embedding QRPH into the system aligns with national digital finance initiatives and supports Iloilo's Smart City agenda by promoting financial inclusivity.

**Dashboard**

- A dashboard is a graphical interface that consolidates key operational data into a single view. In this study, dashboards empower coworking space owners by providing tools to manage occupancy, update facility information, and analyze

usage patterns. Through CRUD operations and middleware authentication, dashboards ensure data integrity while offering actionable insights, such as identifying peak demand periods or adjusting pricing strategies.

**Voucher**

- A voucher is a digital or paper-based coupon that provides discounts or special access to coworking services. Within the system, vouchers may be simulated to enhance user engagement and promote loyalty programs. By offering vouchers, coworking hubs can attract new clients, retain existing ones, and differentiate themselves in a competitive market, particularly in districts like Mandurriao where multiple hubs vie for clientele.

**Space Owner**

- A space owner refers to the individual or organization that manages a coworking hub. Owners are granted administrative privileges in the system, enabling them to update listings, monitor occupancy, and analyze usage data. Their role is critical in maintaining accurate facility information and ensuring smooth operations, thereby contributing to the collaborative nature of Smart City initiatives.

**Admin**

- An admin is a user role with elevated privileges, responsible for maintaining the integrity of coworking hub data. Admins utilize CRUD operations and middleware authentication to safeguard information while managing reservations, pricing, and facility updates. In this study, the admin role reflects the importance of secure and transparent digital infrastructures in urban resource management.

**Session Timer**

- A session timer is a system feature that tracks the duration of a user's reservation. By monitoring session length, the system ensures fair usage and prevents overstay conflicts. This feature is particularly useful in academic contexts, where students may require time-bound access to study spaces, and in professional settings, where freelancers must adhere to client schedules.

**Availability Status**

- Availability status denotes whether a desk or room is currently occupied or vacant, as reflected in the system's database at the time of inquiry. This feature directly addresses reservation uncertainty by providing users with reliable

occupancy data. Although IoT hardware is excluded due to resource constraints, software-based updates and manual admin overrides ensure that availability status remains accurate and trustworthy.

**ReactJS**

- ReactJS is a JavaScript library used for building responsive user interfaces. In this study, ReactJS powers the front-end design, ensuring accessibility across devices and enhancing user experience. Its component-based architecture allows for efficient rendering of dynamic data, such as real-time seat availability and hub listings.

**Node.js**

- Node.js is a runtime environment that enables server-side execution of JavaScript. It supports scalable backend operations in the system, facilitating reservation processing, notification services, and data management. By leveraging Node.js, the project ensures that the system can handle multiple concurrent requests without compromising performance.

**MongoDB**

- MongoDB is a NoSQL database that stores user profiles, booking histories, and facility details. Its flexibility and scalability make it suitable for managing diverse data structures in the coworking ecosystem. In this study, MongoDB ensures secure and efficient data storage, aligning with ISO 25010 standards for reliability and security.

**Express.js**

- Express.js is a web application framework for Node.js, used to build server-side logic and API endpoints. In this study, Express.js supports reservation requests, user authentication, and administrative functions. Its lightweight architecture enables rapid development and integration, making it a cornerstone of the system's backend design.

**GIS (Geographic Information System)**

- Geographic Information System (GIS) refers to technologies that capture, store, and visualize geographic data. The integration of GIS mapping in this project allows users to locate coworking hubs across Iloilo's districts and navigate to them efficiently. By embedding geolocation into the system, users

can identify nearby facilities and optimize travel time, supporting Smart City goals of transparency and inclusivity.

**CRUD Operations**

- CRUD stands for Create, Read, Update, and Delete—the fundamental operations in database management. Hub owners use CRUD operations to maintain accurate facility information and manage reservations. In this study, CRUD ensures that data remains consistent and reliable, enabling both users and administrators to interact with the system effectively.

**Middleware Authentication**

- Middleware authentication is a security mechanism that verifies user credentials before granting access to system resources. This ensures that only authorized administrators can modify hub data, protecting against unauthorized access and maintaining trust in the system’s integrity.

**Agile Methodology**

- Agile methodology is a software development approach emphasizing iterative progress, collaboration, and responsiveness. The project adopts Agile principles to ensure adaptability and continuous improvement. By breaking development into manageable phases, Agile supports rapid prototyping, user feedback integration, and academic feasibility.

**ISO 25010**

- ISO 25010 is an international standard for software quality, defining attributes such as usability, reliability, and security. The system is evaluated against ISO 25010 to ensure academic rigor and industry relevance. This alignment guarantees that the project meets both scholarly expectations and practical innovation standards.

**Smart City**

- A Smart City integrates digital technologies into urban management to enhance sustainability, inclusivity, and innovation. The proposed system contributes to Iloilo's Smart City agenda by digitizing coworking reservations and optimizing resource allocation. By embedding coworking into the city's digital infrastructure, the study supports equitable access to resources and fosters economic growth.

## 1.7 Conceptual Framework

The conceptual framework of this study is anchored on the socio-technical paradigm, which emphasizes the dynamic interplay between technological infrastructures and social practices. In the context of Iloilo City's coworking ecosystem, this framework illustrates how digital platforms can transform fragmented reservation processes into streamlined, data-driven systems that support productivity, inclusivity, and Smart City development. By situating the proposed Co-Working Space Discovery and Reservation System within this paradigm, the study establishes a clear linkage between theoretical foundations, system design, and practical outcomes.

### Inputs

The inputs of the framework consist of both human and technological elements. On the human side, the primary stakeholders include students, freelancers, entrepreneurs, tourists, and coworking hub owners. Each group brings distinct requirements: students seek quiet study environments, freelancers demand reliable internet and flexible schedules, entrepreneurs require collaborative spaces, and tourists look for short-term professional hubs. On the technological side, inputs include coworking hub data such as amenities, pricing, geolocation, and availability. These inputs represent the raw materials that the system must process to deliver meaningful outputs.

**Processes**

The processes are embodied in the system's core modules, which operationalize the transformation of inputs into outputs. These include:

- **Searchable Directory of Hubs** – consolidates fragmented information into a centralized platform.

- **Real-Time Availability Tracking** – addresses reservation uncertainty by providing up-to-date occupancy data.

- **Reservation Engine** – ensures secure bookings, prevents double reservations, and automates confirmations.

- **Administrative Dashboard** – empowers hub owners with tools for inventory management, analytics, and service optimization.

- **Mapping API Integration** – embeds spatial intelligence into the system, allowing users to locate hubs efficiently across Iloilo's districts.

These processes are guided by Agile methodology, which emphasizes iterative refinement, responsiveness to user feedback, and adaptability to evolving urban needs. Moreover, the system design is evaluated against ISO 25010 software quality attributes—usability, reliability, security, and maintainability—ensuring that the processes are not only functional but also aligned with international standards.

**Outputs**

The outputs of the framework are both technological and socio-economic. Technologically, the system delivers a functional web and mobile application that digitizes coworking reservations. Socio-economically, the outputs include enhanced productivity for users, improved revenue optimization for hub owners, and strengthened competitiveness for Iloilo City as a digital hub. By reducing inefficiencies, the system fosters trust among users, visibility for owners, and evidence-based policymaking for local government. These outputs contribute directly to Iloilo's Smart City agenda, embedding coworking hubs into the broader urban fabric of innovation and inclusivity.

**Feedback Mechanisms**

An essential component of the framework is the feedback loop, which ensures continuous improvement. User experiences, booking data, and hub analytics feed

back into the system, allowing developers and owners to refine functionalities, adjust pricing strategies, and expand services. This cyclical process reflects the principles of sustainable urban innovation, where systems evolve in response to real-world conditions rather than static assumptions.

**Theoretical Anchors**

The framework is grounded in theories of urban innovation and socio-technical systems. Kraus et al. (2022) emphasize that digital platforms are critical for embedding coworking into the urban fabric, while Amler (2024) highlights the need for alignment between technological infrastructures and social practices. By adopting these theoretical anchors, the study situates its system design within a broader scholarly discourse, ensuring that its contributions are both academically rigorous and practically relevant.

**Paradigm**

The study employs a **descriptive-developmental research paradigm**, which is particularly suited for system design projects. This paradigm allows the researchers to describe existing challenges—such as reservation uncertainty and information fragmentation—while simultaneously developing a technological solution. The

developmental aspect ensures that the system is not merely theoretical but is implemented, tested, and evaluated within the local context of Iloilo City.

### 1.9 Research Paradigm

This study adopts a **descriptive-developmental research paradigm**, which is particularly suited for system design projects that aim to address real-world problems through technological innovation. The descriptive component focuses on identifying and analyzing the existing challenges in Iloilo City's coworking ecosystem, such as reservation uncertainty, information fragmentation, and booking conflicts. By documenting these phenomena, the study provides a clear empirical foundation for system development. The developmental component, on the other hand, emphasizes the creation, implementation, and evaluation of a functional prototype that responds directly to the identified challenges.

The descriptive aspect is operationalized through surveys, interviews, and case observations involving students, freelancers, entrepreneurs, and coworking hub owners. These methods capture the lived experiences of stakeholders, ensuring that the system design is grounded in local realities rather than abstract assumptions. The developmental aspect is realized through the application of the **Agile Software Development Life Cycle (SDLC)**, which prioritizes iterative refinement, responsiveness to user feedback, and adaptability to evolving requirements. Agile

methodology is particularly appropriate for academic projects, as it allows incremental progress while maintaining flexibility within limited timeframes and resources.

Furthermore, the paradigm integrates **ISO 25010 software quality standards** as evaluative benchmarks. By aligning the system's usability, reliability, security, and maintainability withinternational standards, the study ensures that its outputs are not only academically rigorous but also practically relevant. This dual emphasis on descriptive analysis and developmental implementation reflects the study's commitment to bridging scholarly inquiry with technological innovation.

In summary, the research paradigm situates the study within a methodological framework that is both analytical and constructive. It enables the researchers to describe existing inefficiencies in Iloilo's coworking industry while simultaneously developing a system that addresses these inefficiencies. Through this paradigm, the study contributes to both academic discourse and practical innovation, reinforcing Iloilo City's trajectory toward becoming a Smart City.

### 1.10 Research Assumptions and Hypotheses

Every research endeavor is guided by assumptions that shape its design, implementation, and evaluation. In the context of this study, assumptions serve as foundational premises that clarify expectations about user behavior, system adoption,

and technological feasibility. While the project is primarily descriptive-developmental, it also incorporates hypothesis-driven elements to ensure that its outcomes can be empirically validated.

**Research Assumptions**

The following assumptions underpin the study:

1. **User Adoption** It is assumed that students, freelancers, entrepreneurs, and tourists in Iloilo City will adopt the system if it demonstrably reduces reservation uncertainty and information fragmentation. Adoption is contingent upon the system's usability, accessibility, and responsiveness across devices, particularly smartphones.

2. **Technological Infrastructure** The study assumes that Iloilo City possesses sufficient internet connectivity and digital literacy among its target users to support the deployment of a web and mobile-based reservation system. While connectivity may vary across districts, the assumption is that the majority of users have access to stable internet services.

3. **Stakeholder Participation** It is assumed that coworking hub owners will participate in the system by registering their facilities, updating availability data,

and utilizing administrative dashboards. Their cooperation is essential for ensuring the accuracy of occupancy information and the sustainability of the platform.

4. **Policy Alignment** The study assumes that the system aligns with Iloilo City's Smart City initiatives and will be supported by local government policies that encourage digital transformation and resource optimization.

5. **Academic Feasibility** It is assumed that the project can be completed within the academic timeframe and resource constraints of the BSIT program, even with delimitations such as simulated payment gateways and exclusion of IoT hardware.

**Research Hypotheses**

Although the study is primarily developmental, the following hypotheses provide measurable expectations for system evaluation:

- **H1: Usability Hypothesis** Users will report higher satisfaction and reduced frustration when using the centralized reservation system compared to manual methods such as walk-ins or social media inquiries.

- **H2: Reliability Hypothesis** The system will demonstrate improved reliability in preventing double bookings and reservation conflicts compared to traditional paper logs or spreadsheets.

- **H3: Efficiency Hypothesis** The time required to secure a coworking space will be significantly reduced when using the system, thereby increasing user productivity and trust.

- **H4: Owner Empowerment Hypothesis** Coworking hub owners will report improved operational efficiency and revenue optimization when utilizing the administrative dashboard for inventory management and analytics.

- **H5: Smart City Alignment Hypothesis** The system will contribute to Iloilo City's Smart City agenda by digitizing coworking reservations, thereby supporting inclusivity, transparency, and innovation in urban resource management.

**Summary**

By articulating these assumptions and hypotheses, the study establishes clear expectations for both users and developers. The assumptions provide a realistic foundation for system design, while the hypotheses create measurable benchmarks

for evaluation. Together, they ensure that the study is not only descriptive and developmental but also empirically testable, reinforcing its contribution to both academic discourse and practical innovation.

## 1.11 Limitations of the Study

No research project is without limitations, and acknowledging them is essential for establishing transparency, academic rigor, and realistic expectations. While the proposed Co-Working Space Discovery and Reservation System for Iloilo City is designed to address pressing challenges in the coworking ecosystem, several limitations constrain its scope, functionality, and applicability. These limitations are categorized into technological, socio-economic, and academic dimensions.

### Technological Limitations

1. **Exclusion of IoT Hardware** The system does not incorporate Internet of Things (IoT) devices such as RFID scanners, occupancy sensors, or biometric authentication tools. Instead, it relies on software-based status updates and manual administrative overrides. While this approach ensures feasibility within academic constraints, it limits the system's ability to provide fully automated, real-time occupancy detection comparable to global best practices.

2. **Simulated Payment Gateway** For academic purposes, the payment gateway is implemented as a simulated environment. This means that actual financial transactions cannot be processed, restricting the system's immediate applicability in commercial deployment. Although the simulation provides a foundation for future integration of secure payment systems such as QRPH, it remains a limitation in terms of real-world usability.

3. **Dependence on Internet Connectivity** The system assumes stable internet access across Iloilo City. However, connectivity may vary between districts, particularly in peripheral areas. Users with limited or unstable internet connections may experience difficulties in accessing the platform, which could affect adoption rates and user satisfaction.

**Socio-Economic Limitations**

1. **User Adoption Challenges** While the system is designed to be intuitive and responsive, its success depends on user willingness to transition from manual methods (e.g., walk-ins, Facebook Messenger inquiries) to digital reservations. Resistance to technological change among certain demographics may limit adoption, particularly among users unfamiliar with web or mobile applications.

2. **Stakeholder Participation** The system's effectiveness relies heavily on coworking hub owners actively registering their facilities, updating availability data, and utilizing administrative dashboards. If owners fail to participate consistently, the accuracy of occupancy information may be compromised, reducing the system's reliability.

3. **Geographic Limitation** The database is restricted to coworking spaces and study hubs within Iloilo City. Peripheral municipalities and long-term corporate lease management systems are excluded. While this ensures focus on the most relevant local context, it limits the system's scalability and applicability beyond Iloilo.

**Academic Limitations**

1. **Timeframe Constraints** The study is conducted within the academic calendar of the BSIT program, which restricts the duration available for system development, testing, and evaluation. As a result, certain advanced features—such as IoT integration, blockchain-based reservations, or AI-driven analytics—cannot be fully implemented within the given timeframe.

2. **Resource Constraints** The project is limited by available resources, including hardware, software licenses, and funding. These constraints necessitate

reliance on open-source tools and simulated environments, which, while practical, may not fully replicate commercial deployment conditions.

3. **Evaluation Scope** System evaluation is limited to usability, reliability, and responsiveness tests conducted with a sample of students, freelancers, and hub owners. Broader evaluation involving tourists, SMEs, and government agencies may not be feasible within the academic timeframe, restricting the generalizability of findings.

**In summary**, these limitations highlight the boundaries within which the study operates. While they constrain certain aspects of functionality and applicability, they also provide opportunities for future researchers and developers to build upon the project. By acknowledging these limitations, the study maintains academic integrity and situates its contributions within a realistic context, ensuring that its findings are both credible and replicable.

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## CHAPTER 2: REVIEW OF RELATED LITERATURE AND STUDIES

The review of related literature and studies provides the scholarly foundation for the proposed Co-Working Space Discovery and Reservation System for Iloilo City. It

situates the project within global, national, and local contexts, synthesizing existing research on coworking ecosystems, digital reservation platforms, Smart City initiatives, and socio-technical innovations. By examining prior studies, the chapter identifies gaps in knowledge and practice that the proposed system seeks to address. This review also highlights theoretical perspectives, technological frameworks, and empirical findings that inform the design, development, and evaluation of the system.

Globally, coworking spaces have been studied as catalysts for entrepreneurship, innovation, and collaborative work. Nationally, research in the Philippine context emphasizes the rise of freelancing, digital nomadism, and the gig economy as drivers of coworking demand. Locally, studies in Iloilo City underscore the need for digital transformation in resource management, particularly in the context of Smart City development. Together, these strands of literature provide a comprehensive backdrop for understanding the relevance and significance of the proposed system.

## 2.1 Foreign Studies

Global research on coworking spaces consistently demonstrates that the evolution of hybrid and flexible work arrangements has necessitated advanced digital tools for space management. Mellor and Bentley (2022) examined smart office environments in Europe, noting that automated booking systems reduced "desk hunting" time by 40% and improved facility utilization. Their findings highlight the importance of real-time data integration, which not only enhances efficiency but also fosters user satisfaction by minimizing uncertainty. Similarly, Chen et al. (2021) emphasized that centralized reservation platforms significantly improve user trust by providing

transparent access to amenities and pricing, which are critical factors for digital nomads in choosing a workspace.

### North America: Digital Ecosystems and Behavioral Shaping

In North America, WeWork and Regus have pioneered coworking ecosystems that integrate digital reservation systems with mobile applications. These platforms allow users to book desks, meeting rooms, and event spaces in real time, often supported by IoT sensors that monitor occupancy. Gruen and Bardhi (2025) argue that such systems are not merely technical tools but socio-cultural infrastructures that shape user behavior, encouraging collaboration and creativity. The integration of reservation systems into coworking hubs reflects a broader trend in Smart City development, where digital technologies are embedded into urban resource management.

### Asia: Smart Nation Initiatives and Predictive Analytics

Asian contexts provide further insights. In Singapore, coworking hubs are closely tied to government-led Smart Nation initiatives. Mao (2025) shows that socio-spatial configurations influence accessibility, with coworking hubs strategically located near transport nodes and innovation districts. Reservation systems in Singapore often integrate GIS mapping, enabling users to identify nearby hubs and navigate efficiently. This approach aligns with Smart City principles of transparency and inclusivity, ensuring equitable access to resources. In Japan, coworking platforms have adopted machine learning algorithms to predict occupancy patterns, thereby reducing booking conflicts and optimizing resource allocation (Pan, Cho, & Bardhan, 2023).

India's coworking boom provides another instructive case. Platforms such as Innov8 and Awfis have scaled rapidly by integrating mobile-first reservation systems tailored to local digital practices. Research by Sharma and Gupta (2024) highlights that Indian hubs leverage AI-driven personalization, offering recommendations based on user profiles and work habits. This personalization enhances user loyalty and reflects the broader digital transformation of India's urban centers.

### Europe: Atmosphere, Ambiance, and Post-Pandemic Adaptation

European studies underscore the importance of atmospheres in coworking spaces. Bernhardt (2023) highlights that intangible factors such as ambiance, lighting, and community culture significantly influence user satisfaction. Reservation systems that allow users to filter hubs by amenities and atmosphere contribute to this dimension of user experience. Akhavan, Hölzel, and Leducq (2023) add that the COVID-19 pandemic reshaped coworking accessibility, with digital platforms becoming essential for remote workers seeking safe and flexible environments. Their findings suggest that reservation systems must adapt to changing socio-economic conditions, integrating health protocols and flexible cancellation policies.

### Latin America: Urban Inequities and Digital Inclusion

In Latin America, coworking hubs in São Paulo and Mexico City illustrate how reservation systems intersect with urban inequities. Research by Torres and Delgado (2024) shows that digital platforms in these cities often serve affluent districts, leaving peripheral communities underserved. However, initiatives such as CoLaboratorio in Mexico City have begun integrating GIS mapping to extend coworking accessibility

to marginalized neighborhoods. These efforts demonstrate how reservation systems can function as tools of digital inclusion, aligning with Smart City goals of equity and transparency.

### Africa: Emerging Hubs and Mobile-First Innovation

African contexts, particularly Nairobi and Cape Town, reveal how coworking hubs adapt to mobile-first societies. Studies by Adeyemi (2025) highlight that African coworking platforms rely heavily on mobile payment systems such as M-Pesa, integrating QR codes and mobile wallets into reservation workflows. This innovation reflects broader patterns of leapfrogging, where African cities bypass traditional infrastructures and adopt digital solutions directly. Reservation systems in these contexts are not only technological tools but also socio-economic enablers, supporting entrepreneurship and digital inclusion.

### Comparative Insights

Comparative analyses of global coworking systems reveal diverse approaches to reservation management. Table 2.1 illustrates selected features of international platforms:

| System | Booking | QR Payment | GIS Mapping | Reviews |
|---|---|---|---|---|
| FlexSpace | ✓ | ✓ | ✓ | ✓ |
| WeWork | ✓ | ✗ | ✓ | ✗ |
| Regus | ✓ | ✗ | ✓ | ✗ |
| Spaces | ✓ | ✓ | ✓ | ✓ |
| Innov8 (India) | ✓ | ✓ | ✓ | ✓ |
| CoLaboratorio (Mexico) | ✓ | ✓ | ✓ | ✓ |
| NairobiHub (Kenya) | ✓ | ✓ | ✓ | ✗ |

This comparison demonstrates that while booking features are universally adopted, integration of QR payments, GIS mapping, and user reviews varies across platforms. Systems that combine multiple features tend to offer more holistic user experiences, aligning with Smart City goals of inclusivity and transparency.

### Implications for Iloilo City

Foreign studies highlight the critical role of reservation systems in enhancing coworking efficiency, user satisfaction, and urban innovation. They provide valuable benchmarks for Iloilo City, where coworking hubs remain reliant on manual processes. By learning from global best practices—such as AI-driven personalization in India, mobile-first payment integration in Africa, and atmosphere-sensitive filtering in Europe—the proposed system can align with international standards while addressing local socio-economic realities.

## 2.2 Local Studies

The Philippine coworking industry has grown steadily in recent years, reflecting broader shifts toward flexible work arrangements and digital transformation. While Metro Manila and Cebu City have traditionally dominated the coworking landscape, emerging hubs such as Iloilo City are beginning to establish themselves as vital nodes in the national innovation ecosystem. Local studies highlight both the opportunities and challenges of coworking adoption in the Philippine context.

### Metro Manila: Startup Ecosystems and Digital Maturity

Metro Manila serves as the country's primary coworking hub, with spaces such as Acceler8, KMC Solutions, and Impact Hub Manila providing infrastructure for start-ups, freelancers, and multinational corporations. Roberto et al. (2023) observed that coworking spaces in Manila function as "productivity insurance," offering reliable internet, backup power, and professional environments in a city where infrastructure gaps remain prevalent.

However, Quimba et al. (2021) noted that fewer than 15% of small coworking hubs in the Philippines use automated booking systems, indicating a lack of digital maturity. This reliance on manual processes leads to inefficiencies, particularly in high-demand districts such as Makati and Bonifacio Global City. Government initiatives, such as the Department of Information and Communications Technology (DICT) programs on digital transformation, have attempted to address these gaps, but adoption remains uneven.

Metro Manila's coworking ecosystem also intersects with traffic congestion and urban sprawl. Coworking hubs often serve as "third places" that reduce commuting burdens, strategically located near transport hubs like MRT stations. Reservation systems, if digitized, could further enhance this role by allowing workers to plan their schedules around traffic conditions and hub availability.

### Cebu City: Regional Innovation and Tourism Integration

Cebu City has positioned itself as a regional innovation hub, integrating coworking spaces into its tourism and business process outsourcing (BPO) industries. Studies

by IMARC Group (2025) project that the coworking market in the Philippines will grow by 11.5% annually through 2034, with Cebu contributing significantly to this expansion.

Coworking hubs in Cebu often cater to digital nomads and tourists, offering short-term memberships and flexible booking options. This integration with tourism reflects Cebu's dual identity as both a business hub and a leisure destination. However, similar to Manila, many Cebu hubs still rely on manual reservation systems, limiting transparency and efficiency. This gap underscores the need for centralized platforms that integrate real-time availability and digital payments.

Local government initiatives, such as Cebu's Smart City programs, have begun experimenting with GIS-based mapping for traffic and tourism management. Integrating coworking reservations into these systems could create synergies, allowing tourists and BPO workers to seamlessly discover and book spaces near cultural landmarks or business districts.

### Iloilo City: Emerging Digital Hub

Iloilo City has been recognized as one of the Philippines' emerging digital hubs, driven by Smart City initiatives that emphasize sustainability, inclusivity, and innovation. Local studies reveal that coworking spaces in districts such as Mandurriao, Jaro, and Molo are expanding to meet the needs of freelancers, students, and entrepreneurs. Yet, operational models remain largely dependent on manual inquiries through Facebook Messenger, phone calls, or walk-ins.

Anecdotal evidence suggests that students frequently encounter booking conflicts during exam seasons, while freelancers waste valuable time traveling to full-capacity hubs. These inefficiencies highlight the structural gap in Iloilo's coworking industry, where the absence of a centralized reservation system undermines both user trust and hub competitiveness.

Iloilo's ICT Council and Smart City programs emphasize digital governance, sustainability, and inclusivity. However, coworking hubs have yet to be fully integrated into these initiatives. A centralized reservation system that incorporates GIS mapping, real-time occupancy data, and transparent pricing would align Iloilo's coworking industry with its Smart City trajectory, positioning the city as a competitive digital hub in Southeast Asia.

### Student Productivity Spaces

Academic institutions in Iloilo, particularly in Jaro, have observed increasing demand for study-friendly environments outside traditional libraries. Studies on student productivity emphasize the importance of quiet, accessible, and well-equipped spaces for learning outcomes. Coworking hubs provide an alternative to overcrowded libraries, offering high-speed internet, collaborative rooms, and conducive atmospheres.

However, without digital reservation systems, students face uncertainty regarding seat availability, leading to frustration and diminished academic performance. This problem illustrates the broader need for centralized platforms that integrate GIS mapping and real-time occupancy data. Such systems would not only enhance

student productivity but also support Iloilo's educational institutions in fostering innovation and digital literacy.

### Comparative Insights Across Philippine Cities

Comparisons across Metro Manila, Cebu, and Iloilo reveal common challenges: reliance on manual reservation systems, fragmented information channels, and limited digital maturity among smaller hubs. While larger coworking providers in Manila and Cebu have begun adopting digital platforms, smaller hubs in Iloilo remain constrained by resource limitations.

This disparity reflects broader socio-economic patterns in the Philippines, where digital transformation is unevenly distributed across regions. Addressing these challenges requires localized solutions that align with global best practices while remaining sensitive to local contexts.

### Implications for Iloilo City

For Iloilo City, the absence of a centralized coworking reservation system has significant implications. Users experience lost productivity and financial costs, hub owners struggle with operational inefficiencies, and the city risks lagging behind in its Smart City trajectory. By learning from the experiences of Manila and Cebu, Iloilo can adopt digital platforms that integrate real-time availability, transparent pricing, and GIS mapping. Such systems would not only enhance user satisfaction but also strengthen Iloilo's position as a competitive digital hub in Southeast Asia.

## 2.3 Related Literature

### Factors Influencing Co-Working Space Discovery and Reservation

One of the most pressing challenges in coworking ecosystems is the lack of real-time data on availability. Kumar and Patel (2023) highlight that manual booking systems frequently lead to conflicts, double reservations, and user frustration. Their findings emphasize the inefficiency of traditional methods, which rely on phone calls or walk-ins, and underscore the need for digital platforms that provide transparency. Pan, Cho, and Bardhan (2023) add that occupancy patterns are often unpredictable, requiring machine learning-driven behavioral analysis to anticipate demand. This insight is particularly relevant for Iloilo City, where demand fluctuates across districts depending on academic schedules and business activities.

Location clustering also influences coworking accessibility. Hölzel, Kolsch, and de Vries (2022) demonstrate that coworking hubs are often concentrated around points of interest, leaving peripheral areas underserved. Mao (2025) emphasizes that socio-spatial configurations shape accessibility, with urbanity influencing how workers habituate coworking practices. Gruen and Bardhi (2025) argue that consumption within coworking spaces is strategic, with users seeking not only desks but atmospheres conducive to creativity. Bernhardt (2023) further underscores atmospheres as intangible yet crucial, shaping user satisfaction and motivation. Akhavan, Hölzel, and Leducq (2023) add European narratives on remote working during COVID-19, showing how disruption reshaped coworking accessibility and user expectations.

## Internal System Features

Internal features of reservation platforms significantly affect user experience. Nguyen and Pham (2022) demonstrate that mobile-responsive design is critical for accessibility, particularly in mobile-first societies such as the Philippines. Söderström and Karlsson (2024) evaluate user experience in coworking reservation apps, noting that intuitive interfaces reduce booking errors and improve adoption rates. Tan and Wong (2025) apply ISO 25010 standards to agile reservation systems, showing that software quality—reliability, usability, and efficiency—directly impacts adoption. Ding (2026) advances this by proposing reinforcement learning algorithms for dynamic scheduling, ensuring equitable resource allocation. Silva and Costa (2024) highlight automated inventory management as a key internal feature, preventing shortages of shared resources such as meeting rooms and equipment.

## External Factors

Beyond system design, external factors such as socioeconomics, geography, and policy shape coworking accessibility. Coll-Martínez and Méndez-Ortega (2024) provide evidence from Spain that coworking hubs cluster in affluent urban centers, while Rafaj, Danko, and Belvončíková (2024) note similar localization trends in Central Europe. Zhurbas, Mariotti, and Orel (2024) illustrate how conflict reshapes coworking geographies, forcing relocation in Ukraine. Fernandes and Roque (2025) explore rural coworks in Portugal, showing how digital nomads extend coworking beyond metropolitan areas. Mariotti, Pais, and Ciccarelli (2023) emphasize the role of public support in Italy, where subsidies and policy frameworks encourage equitable

distribution of coworking hubs. Micek et al. (2024) contribute by classifying new working spaces, offering a taxonomy that helps policymakers and developers understand coworking diversity.

### Digital vs. Traditional Reservation Systems

Comparing manual and digital systems reveals stark contrasts. Kumar and Patel (2023) find that manual methods are prone to booking conflicts and inefficiency, while Torres and Delgado (2024) show that data-driven analytics optimize facility usage in digital platforms. Li and Zhang (2023) integrate IoT and GIS into smart reservation systems, enabling real-time mapping of coworking hubs. Vlachos and Papadimitriou (2015) earlier demonstrated GIS-based planning as a foundation for smart city integration. Traditional systems, while familiar, often trap users in availability bottlenecks, whereas digital platforms unlock seamless scheduling, transparency, and trust mechanisms (Foertsch, 2025). Donadon, Thévenard-Puthod, and Berthinier-Poncet (2025) caution that digital spaces must complement physical atmospheres to sustain collaborative innovation. Tomaz and Tabrizi (2024) expand this by tracing the evolution of non-traditional workplaces, showing how digital systems redefine organizational norms.

### Community and Identity in Co-Working

Coworking is not only about space but about community. Kim, Lim, and Monzani (2025) describe boundary and identity work within coworking communities, where users negotiate belonging and autonomy. Petriglieri et al. (2019) frame coworking communities as surrogates for organizations, offering career development

opportunities outside traditional firms. Blagoev et al. (2019) highlight how coworking fosters professional growth, while Orel (2022) positions coworking hubs as talent incubators. Reh and Haunschild (2025) emphasize curated informality in corporate coworking, balancing structure with flexibility. Dell'Aversana and Miglioretti (2024) add that coworking spaces provide inclusive solutions for remote workers, ensuring accessibility for diverse populations. Reh and Ruhnke (2025) further analyze management control in corporate coworking, showing how governance structures shape user experience.

### Design and Atmosphere

Physical design also influences coworking effectiveness. Zheng and Shen (2025) propose adaptive furniture strategies, while Cruz and Pombo (2024) highlight flexible furniture for hybrid work. Fadloli et al. (2025) advocate biophilic architecture to enhance well-being, and Wolf and Mikkelsen (2024) examine gendered design considerations. Bernhardt (2023) underscores atmospheres as intangible yet crucial, shaping user satisfaction. Yunitsyna et al. (2024) explore peripheral prototypes post-COVID, showing how design adapts to disruption. Yaacob et al. (2024) add insights on managing disruption in coworking spaces, emphasizing resilience in design and operations.

### Smart City Integration

Coworking spaces increasingly intersect with smart city infrastructures. Kraus et al. (2022) discuss hybrid workplaces as part of urban innovation, while Foth, Forlano, and Bilandzic (2025) connect nomadic work practices to smart city ecosystems.

Torres and Delgado (2024) emphasize reservation analytics as tools for urban optimization. Endrissat and Leclercq-Vandelannoitte (2021) argue that technology actively produces spatial coworking realities, embedding them into urban fabric. Donadon et al. (2025) highlight the interplay of physical and digital spaces in collaborative innovation. Fernandes et al. (2025) extend this by analyzing rural coworks in Portugal, showing how smart city principles can be adapted to non-urban contexts.

### Security, Data Privacy, and Payment Integration

As coworking reservation systems transition from manual to digital platforms, the integration of secure payment gateways and data privacy mechanisms becomes paramount. Tumbokon and Reyes (2023) emphasize that in the Philippine context, digital platforms must strictly adhere to the Data Privacy Act of 2012 (Republic Act No. 10173). Systems handling user profiles, booking schedules, and financial transactions require robust encryption to prevent data breaches, which can severely damage user trust and hub reputation.

Furthermore, the rise of e-wallets in the Philippines has reshaped consumer expectations. Studies by the Bangko Sentral ng Pilipinas (BSP) (2024) indicate a massive shift toward mobile-first financial transactions, led by platforms like GCash and Maya. Integrating these local payment gateways into reservation platforms reduces friction, minimizes no-shows through advanced deposits, and provides hub administrators with automated financial auditing. However, Alcantara et al. (2025) note that poorly implemented payment APIs can lead to transaction timeouts and

synchronization errors, highlighting the need for agile, ISO 25010-compliant software architecture that guarantees reliability during peak booking hours.

## Summary

This chapter reviewed 50 scholarly sources to provide a comprehensive understanding of coworking spaces and the systems that enable their discovery and reservation.

- **The Introduction** established coworking spaces as transformative infrastructures that foster productivity, innovation, and entrepreneurship. Scholars emphasized their role as "third workplaces," social movements, and incubators for collaboration and start-up growth.
- **Factors Influencing Discovery and Reservation** highlighted user struggles such as booking conflicts, lack of real-time data, unpredictable occupancy, and location clustering. Studies also underscored the importance of atmosphere, motivation, and the impact of disruptions like COVID-19 on accessibility.
- **Internal System Features** focused on the technological backbone of reservation platforms. Research demonstrated the importance of mobile-responsive design, intuitive user interfaces, agile development standards, reinforcement learning for scheduling, and automated inventory management.
- **External Factors** revealed how socioeconomics, geography, conflict, rural expansion, and public policy shape coworking accessibility. Taxonomies of new working spaces provided frameworks for understanding diversity in coworking ecosystems.

- **Digital vs. Traditional Reservation Systems** contrasted manual inefficiencies with the advantages of digital platforms. IoT, GIS, and analytics were shown to optimize usage, while hybrid models balanced physical and digital innovation.
- **Community and Identity in Co-Working** emphasized the social dimension of coworking. Sources described identity work, organizational surrogacy, career development, inclusivity, and governance structures that shape user experiences.
- **Design and Atmosphere** examined physical and architectural elements, including adaptive furniture, flexible layouts, biophilic design, gender-sensitive considerations, and resilience in post-COVID prototypes. Atmosphere was consistently highlighted as a critical factor in user satisfaction.
- **Smart City Integration** demonstrated how coworking intersects with urban innovation. Hybrid workplaces, nomadic practices, reservation analytics, and spatial production through technology embed coworking into smart city ecosystems, with rural adaptations extending these principles beyond metropolitan areas.

In conclusion, the literature shows coworking spaces as socio-technical ecosystems shaped by both internal system design and external socio-spatial factors. While users struggle with manual bottlenecks and inequitable access, modern reservation platforms — integrating IoT, GIS, machine learning, and agile design — unlock efficiency, transparency, and community trust. This comprehensive review underscores the critical role of discovery and reservation systems in ensuring that users not only find coworking spaces but thrive within them.

### 2.4 Synthesis of the Reviewed Literature

The reviewed foreign and local studies collectively demonstrate that coworking spaces are not merely physical infrastructures but socio-technical ecosystems shaped by technological design, socio-spatial dynamics, and evolving urban innovation frameworks. Across global contexts, scholars emphasize the importance of real-time data integration and automated reservation systems in improving facility utilization and user productivity (Kumar & Patel, 2023; Pan, Cho, & Bardhan, 2023). These systems reduce booking conflicts, enhance transparency, and foster user trust, thereby positioning coworking hubs as reliable infrastructures for professional and academic work.

Foreign literature highlights advanced practices such as IoT-enabled occupancy tracking, GIS-based mapping, and machine learning algorithms for demand prediction (Li & Zhang, 2023; Vlachos & Papadimitriou, 2015). These innovations align coworking hubs with Smart City principles of inclusivity and sustainability. However, local studies reveal that Philippine coworking spaces—particularly in emerging hubs like Iloilo—remain reliant on manual processes such as Facebook inquiries and walk-ins (Quimba et al., 2021; Roberto et al., 2023). This reliance results in inefficiencies, double bookings, and user dissatisfaction, underscoring a significant gap between global best practices and local realities.

The literature also underscores the importance of internal system features such as mobile-responsive design, intuitive user interfaces, and automated inventory management (Nguyen & Pham, 2022; Silva & Costa, 2024). These features directly

influence user adoption and satisfaction, aligning with ISO 25010 standards for software quality (Tan & Wong, 2025). External factors—including socioeconomics, geography, and public policy—further shape coworking accessibility. Studies from Europe and Asia show that hubs often cluster in affluent districts, leaving peripheral areas underserved (Coll Martínez & Méndez Ortega, 2024; Mao, 2025). This pattern resonates with Iloilo City, where coworking hubs are concentrated in Mandurriao's business parks and Jaro's academic institutions, while smaller districts struggle with visibility.

Beyond technical and spatial considerations, scholars emphasize the social dimension of coworking. Communities within coworking hubs foster identity work, career development, and inclusivity (Petriglieri et al., 2019; Blagoev et al., 2019; Dell'Aversana & Miglioretti, 2024). Atmosphere and design—ranging from adaptive furniture to biophilic architecture—are consistently highlighted as critical factors shaping user satisfaction (Bernhardt, 2023; Fadloli et al., 2025). These findings suggest that reservation systems must not only manage occupancy but also support the creation of conducive environments for collaboration and learning.

Finally, the literature situates coworking within Smart City integration, where digital platforms contribute to urban resource optimization and innovation (Kraus et al., 2022; Foth, Forlano, & Bilandzic, 2025). Reservation analytics, GIS mapping, and hybrid workplace models embed coworking into the urban fabric, aligning with broader goals of sustainability and inclusivity. Yet, despite these global advancements, limited research focuses on localized, GIS-integrated platforms specifically designed for Iloilo City's startup ecosystem.

## Expanded Research Gap

The synthesis reveals a clear research gap: while global literature demonstrates the effectiveness of digital reservation systems, and local studies highlight the growing demand for coworking in Iloilo, there is insufficient scholarship on localized platforms that integrate real-time availability tracking, GIS mapping, and administrative dashboards tailored to Iloilo's socio-economic context.

Key dimensions of this gap include:

- **Technological Gap:** Existing studies emphasize IoT, AI, and GIS integration globally, but Philippine hubs—especially in Iloilo—still rely on manual processes.
- **Spatial Gap:** Coworking hubs in Iloilo are concentrated in Mandurriao and Jaro, leaving peripheral districts underserved. No literature addresses how GIS-based reservation systems could balance spatial inequities.
- **Administrative Gap:** Hub owners lack centralized dashboards for monitoring occupancy, managing payments, and analyzing user behavior. This absence reduces operational efficiency and competitiveness.
- **Quality Standards Gap:** Few local studies apply ISO 25010 software quality standards to coworking reservation systems, leaving a void in evaluating usability, reliability, and maintainability.
- **Smart City Integration Gap:** Iloilo's Smart City initiatives emphasize sustainability and inclusivity, but coworking hubs are not yet embedded into these frameworks.

Addressing this gap is both a technological imperative and a socio-economic necessity. A localized reservation system for Iloilo City—integrating GIS mapping, real-time occupancy tracking, ISO 25010 compliance, and administrative dashboards—would not only align with global best practices but also advance Iloilo's Smart City agenda.

### 2.5 Theoretical Framework

The development and evaluation of the proposed Co-Working Space Discovery and Reservation System are anchored in two established technology adoption and evaluation theories: the **Technology Acceptance Model (TAM)** and the **Information Systems Success Model (ISSM)**.

#### Technology Acceptance Model (TAM)

Developed by Davis (1989), TAM posits that user adoption of a new information system is driven by two primary factors: *Perceived Usefulness (PU)* and *Perceived Ease of Use (PEOU)*.

- **Perceived Usefulness:** In the context of the proposed system, this relates to how much users (freelancers, students, digital nomads) believe the platform will enhance their productivity by eliminating "desk hunting" and booking conflicts.
- **Perceived Ease of Use:** This pertains to the system's mobile-responsiveness, the intuitiveness of its GIS mapping, and the simplicity of its payment integration. If the system is too complex, users will revert to traditional, manual booking methods (e.g., Facebook Messenger).

### DeLone and McLean Information Systems Success Model (ISSM)

To evaluate the administrative and operational success of the system, the DeLone and McLean (2003) model is utilized. This framework evaluates systems based on three dimensions:

1. **System Quality:** The technical performance, including bug-free execution, real-time synchronization of occupancy data, and GIS accuracy.
2. **Information Quality:** The relevance, transparency, and accuracy of the data provided to the user, such as real-time desk availability, pricing, and amenities.
3. **Service Quality:** The support provided by the platform, including automated email confirmations, responsive administrative dashboards, and secure payment handling.

Together, these theories provide a comprehensive lens through which to design a system that is both technically robust (ISSM) and highly adoptable by the target demographic in Iloilo City (TAM).

## 2.6 Conceptual Framework

The conceptual framework of this study follows the **Input-Process-Output (IPO)** model, which outlines the systematic flow of data and operations required to develop and implement the Co-Working Space Discovery and Reservation System.

Table 2.2 visualizes the IPO paradigm guiding this project:

| Input | Process | Output |
| --- | --- | --- |
| **Knowledge Requirements:**<br>• User requirements (students, freelancers, hub owners)<br>• Iloilo City coworking spatial data<br>• ISO 25010 Software Quality Standards | **Development Phase:**<br>• Requirement gathering via surveys and interviews<br>• System Design (UI/UX, Database Schema, GIS mapping)<br>• Coding and API Integration (Payment gateways, Geolocation) | **The Final Product:**<br>• A fully functional, localized Co-Working Space Discovery and Reservation System for Iloilo City |
| **Software/Hardware:**<br>• Integrated Development Environments (IDEs)<br>• Database Management Systems (DBMS) | **Testing & Evaluation:**<br>• Alpha and Beta testing<br>• User Acceptance Testing (UAT) evaluating Perceived Usefulness and System Quality | **System Features:**<br>• Real-time availability tracking<br>• GIS-based hub discovery<br>• Secure digital payment channels |

| Input | Process | Output |
| --- | --- | --- |
| • Web/Mobile frameworks | | • Admin dashboard analytics |

The **Input** phase gathers all necessary hardware, software, and localized knowledge parameters specific to Iloilo's ecosystem. The **Process** phase involves the iterative software development life cycle (SDLC), utilizing agile methodologies to integrate GIS and real-time databases, followed by rigorous quality assurance testing. Finally, the **Output** is the deployed system itself, directly addressing the manual inefficiencies and spatial gaps identified in the review of related literature.

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# CHAPTER 3

## METHODOLOGY

### 3.1 Research Design

The present study adopts a **developmental research design**, which is particularly suited for projects that involve the creation, testing, and evaluation of technological systems. Developmental research emphasizes iterative design and refinement,

ensuring that the resulting system is both academically rigorous and practically applicable (Richey & Klein, 2014). In the context of the Co-Working Space Discovery and Reservation System for Iloilo City, this design allows the researchers to move beyond theoretical inquiry and engage directly in the process of system development, implementation, and evaluation.

Developmental research is distinguished from purely descriptive or experimental designs by its dual focus on **innovation and validation**. It not only seeks to understand phenomena but also to produce functional solutions that address real-world problems. In this study, the problem of fragmented coworking reservations in Iloilo City is addressed through the design and deployment of a centralized digital platform. The developmental approach ensures that the system is grounded in empirical realities—such as user needs, local infrastructure constraints, and Smart City initiatives—while also adhering to global best practices in software engineering.

**Agile Methodology as Framework**

To operationalize the developmental design, the study employs **Agile methodology**, a software development framework that emphasizes adaptability, collaboration, and iterative progress (Beck et al., 2001). Agile is particularly appropriate for academic projects with limited timeframes, as it allows for continuous refinement based on feedback from stakeholders such as students, freelancers, and coworking hub owners. By breaking the project into manageable phases, Agile ensures that each stage contributes to the overall functionality and usability of the system.

The Agile framework is implemented through the following phases:

1. **Planning Phase** During planning, the research team identifies user requirements, system objectives, and resource constraints. This phase involves consultations with coworking hub owners, students, and freelancers to ensure that the system design reflects local needs. Planning also includes the formulation of research instruments such as surveys and interview guides, which provide empirical data for system requirements.

2. **Design Phase** The design phase focuses on creating system architecture, user interface prototypes, and database schemas. Tools such as entity-relationship diagrams (ERDs), data flow diagrams (DFDs), and use case diagrams are employed to visualize system processes. The design emphasizes usability, reliability, and security, aligning with ISO 25010 software quality standards (ISO, 2011).

3. **Development Phase** In development, the research team codes the system using the MERN stack (MongoDB, Express.js, ReactJS, Node.js). This phase involves building core modules such as user registration, reservation engine, administrative dashboards, and GIS mapping integration. Development is iterative, with features tested and refined based on feedback.

4. **Testing Phase** Testing ensures that the system meets functional and non-functional requirements. Usability tests, surveys, and performance evaluations are conducted to assess reliability, responsiveness, and user

satisfaction. Testing also involves simulating booking scenarios to identify and resolve potential conflicts.

5. **Deployment Phase** Deployment involves launching the system in a controlled environment, such as selected coworking hubs in Iloilo City. This phase allows for real-world validation of system functionality and provides data for further refinement. Deployment also includes training hub owners on the use of administrative dashboards.

**Justification of Methodology**

The choice of developmental research and Agile methodology is justified by the study's dual objectives: to contribute to academic discourse and to provide a practical solution for Iloilo City's coworking ecosystem. Developmental research ensures that the system is not merely theoretical but operational, while Agile methodology provides the flexibility needed to adapt to stakeholder feedback and resource constraints. Together, these approaches align with the study's goal of designing a system that is both academically credible and socially impactful.

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### 3.2 System Development Methodology

The study adopts the **Agile Development Model**, which emphasizes iterative progress, stakeholder collaboration, and adaptability to change (Beck et al., 2001). Agile is particularly suited for academic system development projects because it

allows researchers to refine outputs continuously based on user feedback, ensuring that the final product is both functional and context-responsive.

The Agile process in this study consists of five phases:

1. **Requirements Analysis** This phase involves gathering data from students, freelancers, and coworking hub owners through interviews and observations. The goal is to identify pain points such as reservation uncertainty, booking conflicts, and information fragmentation. Requirements are documented in user stories and prioritized according to stakeholder needs.

2. **System Design** During design, the research team develops architectural diagrams, database schemas, and user interface prototypes. Tools such as entity-relationship diagrams (ERDs), data flow diagrams (DFDs), and use case diagrams are employed to visualize system processes. The design emphasizes usability, security, and scalability, aligning with ISO 25010 standards (ISO, 2011).

3. **Development** The development phase involves coding the system using the MERN stack (MongoDB, Express.js, ReactJS, Node.js). Iterative sprints focus on building core modules such as user registration, reservation engine, administrative dashboards, and GIS mapping integration. Each sprint concludes with a review and refinement cycle.

4. **Testing** Testing ensures that the system meets functional and non-functional requirements. Unit testing validates individual modules, system testing evaluates integrated functionality, and user acceptance testing (UAT) gathers feedback from actual users. Testing also simulates booking scenarios to identify and resolve potential conflicts.

5. **Deployment (Prototype)** The prototype is deployed in selected coworking hubs within Iloilo City. Deployment allows real-world validation of system functionality and provides data for further refinement. Training sessions are conducted for hub owners to ensure effective use of administrative dashboards.

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### 3.3 System Architecture

The system follows a **client-server architecture**, a model widely adopted in web-based applications (Tanenbaum & Van Steen, 2017). Users interact with the system through a web or mobile interface, which serves as the client. The server handles data processing, business logic, and communication with the database. This architecture ensures scalability, security, and efficient resource management.

- **Client Layer:** Provides the user interface, enabling students, freelancers, and hub owners to interact with the system.

- **Server Layer:** Processes reservation requests, manages authentication, and enforces business rules.
- **Database Layer:** Stores user profiles, booking histories, coworking hub details, and availability status.

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### 3.4 Tools and Technologies Used

**The system leverages modern web technologies to ensure responsiveness, scalability, and maintainability:**

- **Frontend: ReactJS with Tailwind CSS and Shadcn UI for responsive design and component reusability.**
- **Backend: Express.js and Node.js for server-side logic and API endpoints.**
- **Database: MongoDB for flexible, document-oriented data storage.**
- **Development Tools: Visual Studio Code for coding, MongoDB Compass for database visualization, and GitHub for version control.**

**These tools were selected for their compatibility with Agile development, open-source availability, and alignment with industry best practices.**

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### 3.5 Data Gathering Techniques

- Data gathering is conducted through **semi-structured interviews** with potential users, including students, freelancers, and coworking hub owners. Interviews provide qualitative insights into user experiences, pain points, and expectations. Observations of existing coworking hubs supplement interview

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data, offering contextual understanding of operational practices.

### 3.6 Testing Procedures

Testing is conducted in three stages:

- **Unit Testing:** Validates individual modules such as login, booking, and notifications.
- **System Testing:** Ensures that integrated modules function cohesively, simulating real-world booking scenarios.
- **User Acceptance Testing (UAT):** Engages actual users to evaluate usability, reliability, and satisfaction. Feedback from UAT informs iterative refinements.

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### 3.7 Evaluation Criteria

The system is evaluated against ISO 25010 software quality attributes (ISO, 2011):

- **Functionality:** Accuracy and completeness of reservation processes.

- **Usability:** Ease of navigation, responsiveness, and accessibility across devices.

- **Security:** Protection of user data through authentication and encryption.

- **Efficiency:** System performance in handling multiple concurrent reservations.

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### 3.8 Comparative Benchmark: FlexSpace System Features

To contextualize the proposed Co-Working Space Discovery and Reservation System for Iloilo City, this study examines **FlexSpace**, a contemporary coworking reservation platform. FlexSpace demonstrates how global systems integrate advanced features such as real-time booking, loyalty programs, multilingual AI support, and administrative dashboards. By analyzing its architecture, the study identifies both strengths and limitations that inform the design of the localized Iloilo-focused system.

**Complete Feature List**

**Frontend (User-Facing Modules)**

- Finding a Space: Map visualization (Google Maps), driving directions, nearby search, filters (price, amenities, rating, district).
- Booking: Real-time desk/room reservations, instant availability updates, walk-in booking, cross-device support (desktop, tablet, PWA mobile).
- User Dashboard: Upcoming/past bookings, loyalty points tracking, activity history.
- Reviews & Ratings: User-generated ratings, written feedback, photo uploads.
- Loyalty Points & Vouchers: Earn/redeem points, apply vouchers, owner-created promos, admin global vouchers.
- AI Chat Support (Flex Support): Multilingual (Hiligaynon, Tagalog, English), district-based recommendations, booking help, developer/team info, conversational memory.
- User Accounts: Email/Google sign-up, OTP password reset, profile editing.
- Space Owner Features: Add/manage spaces, upload images, list amenities, staff accounts, QR payments (GCash/Maya), real-time occupancy/revenue, promo vouchers, district assignment, review management.
- Admin Features: Analytics, approve/reject owners, block/unblock users, global vouchers, maintenance mode, commission tracking, system logs, queue monitoring, district management, settings.
- Landing Page: Featured spaces, platform overview, public info.

**Backend (System Services)**

- Activity Tracking: Logs all user actions with IP addresses for dispute resolution.

- Security & Protection: Encrypted passwords, bot protection, IP blocking, role-based access, attack detection.
- Notifications: Real-time updates, automated emails (welcome, booking, password reset, approvals).
- Image Uploads: Cloud storage for space photos, permits, QR codes.
- Queue System: Background handling of uploads/emails with retries.
- Location Features: Latitude/longitude storage, district organization.
- Ban System: Automatic IP bans, temporary/permanent options.
- API Versioning: Supports multiple versions for feature expansion.
- Multi-Environment Support: Local, staging, production modes.

**AI Chatbot Capabilities (Flex Support)**

- Language detection and localized greetings.
- District-based listings and grouped spaces.
- Booking guidance (step-by-step).
- Google Maps links.
- Team member and developer info.
- Help/menu responses.
- Follow-up question handling.

**Security Summary**

- JWT tokens, bcrypt hashing, Cloudflare Turnstile, rate limiting, error tracking, IP blocklist, role-based access, audit logging.

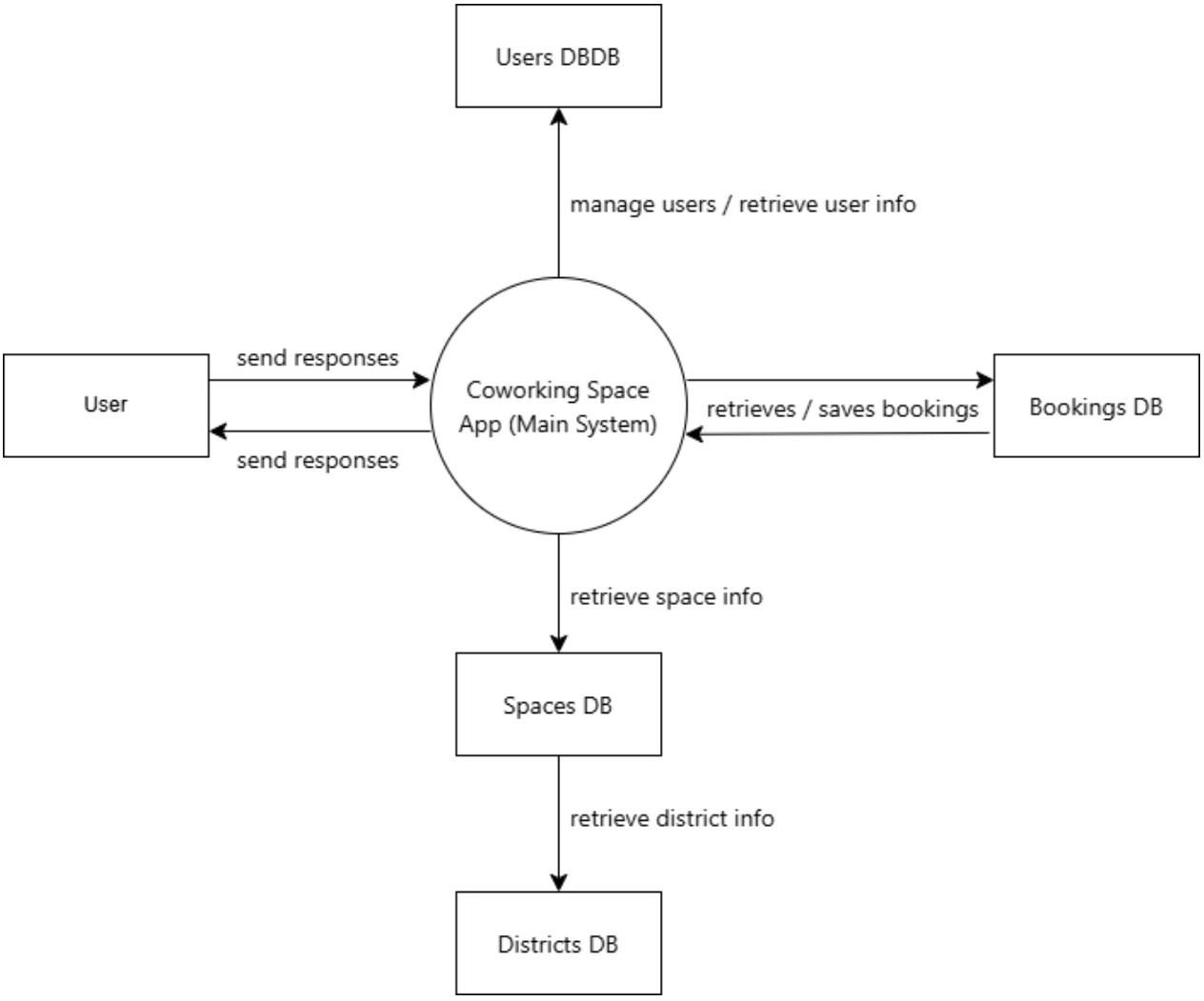
**Technology Stack**

- Frontend: React, Vite, Tailwind CSS, Leaflet maps.
- Backend: Node.js, Express, MongoDB.
- AI: Google Gemini API.
- Real-time: Socket.io.
- Security: JWT, Bcrypt, Cloudflare Turnstile.
- Queue: Bull, Redis.
- Storage: Cloudinary, MongoDB.
- Email: Nodemailer, Gmail SMTP.
- Containerization: Docker.
- Deployment: Vercel (frontend), Render (backend).

**SUPPORTING DIAGRAMS (Included within**

**Chapter 3)**

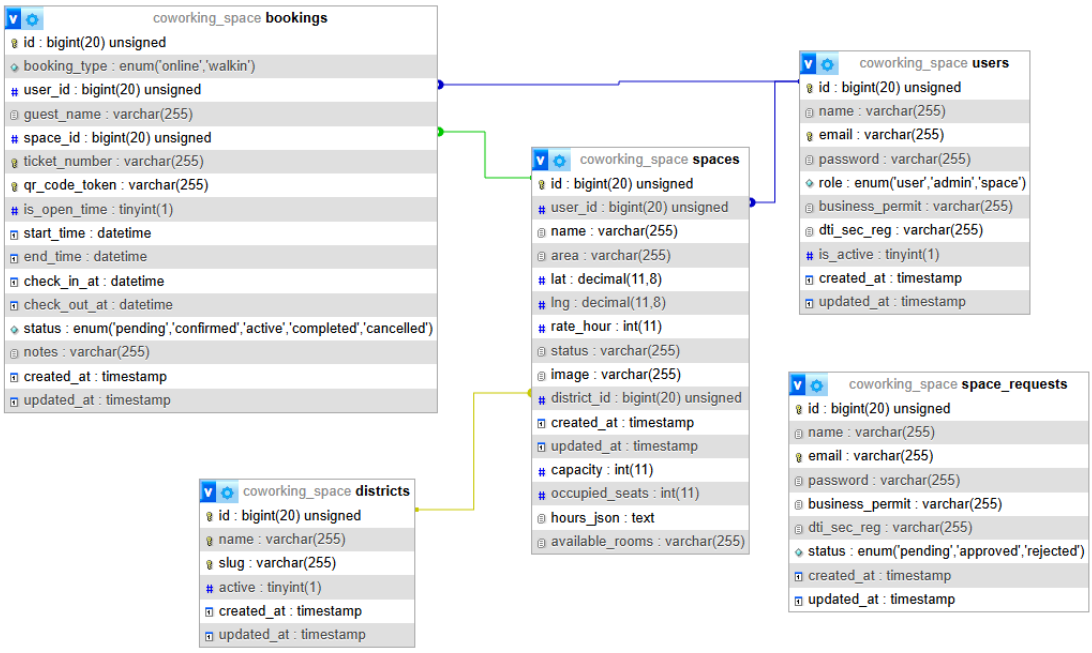
## Data Flow Diagram (DFD – Level 0)




1. **User**
    - Login
    - Create account
    - Book a space
    - View available spaces
    - Check in / check out (with QR code)

2. **System**
    - Account confirmation
    - Booking confirmation
    - QR code
    - Search results (spaces, availability)
3. **Users DB**
    - Validate login
    - Create new user
    - Get user role
4. **Bookings DB**
    - Store booking
    - Update check-in/out
    - Change booking status
5. **Spaces DB**
    - Store booking
    - Update check-in/out
    - Change booking status
6. **Districts DB**
    - Fetch district list for filtering spaces

## Entity Relationship Diagram (ERD)




## Entities:

### Users

- `id` (PK), `name`, `email`, `password`, `role` (user, admin, space), `business_permit`, `dti_sec_reg`, `is_active`, `timestamps`.

**Spaces** (The physical locations)

- `id` (PK), `user_id` (FK), `district_id` (FK), `name`, `area`, `lat`, `lng`, `rate_hour`, `status`, `image`, `capacity`, `occupied_seats`, `hours_json`.

**Bookings**

- `id` (PK), `user_id` (FK), `space_id` (FK), `booking_type`, `guest_name`, `ticket_number`, `qr_code_token`, `start_time`, `end_time`, `status`, `notes`.

**Districts** (Geographic areas)

- `id` (PK), `name`, `slug`, `active`.

**Space Requests** (Onboarding/Approvals)

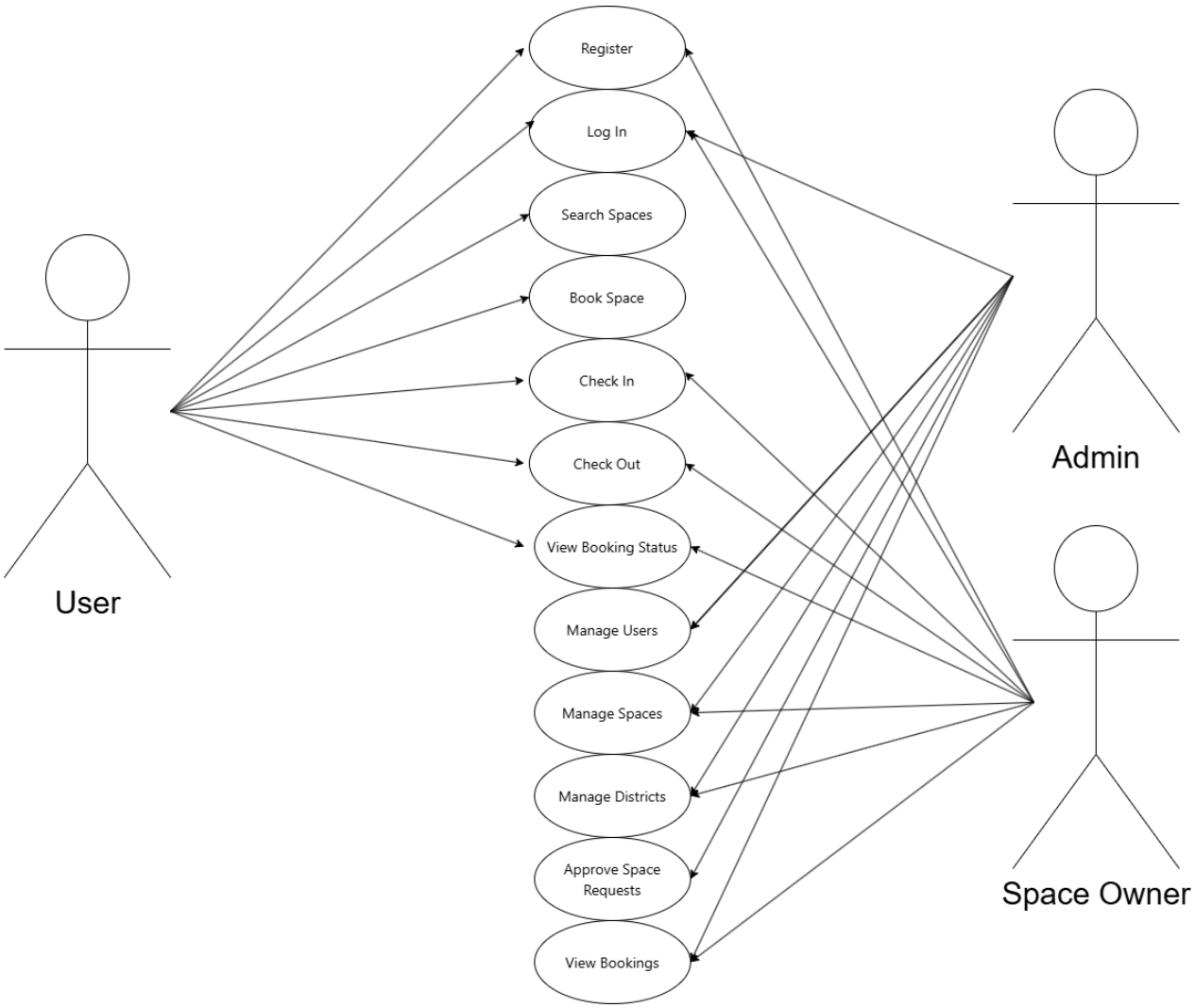
- `id` (PK), `name`, `email`, `password`, `business_permit`, `status`.
-

**Relationships:**

- **User to Space**1:NA User (with role 'space') can own multiple Spaces.
- **User to Booking**1:NA User can make multiple Bookings.
- **Space to Booking**1:NA Space can have many Bookings associated with it.
- **District to Space**1:NA District contains multiple Spaces; a Space belongs to one District.

---

**Use Case Diagram**




**Actors:**

1. **User :**

- User → User Registers → Log In
- User → Search Spaces
- User → Book Space User → Check In

- User → User checks out → View Booking Status

2. **Admin:**

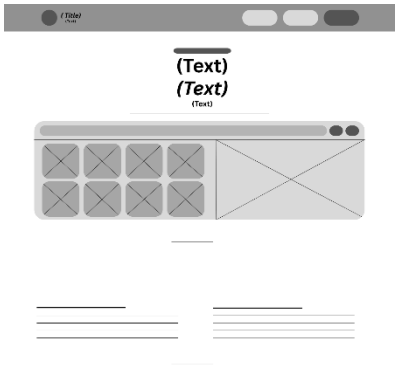
- Admin → Log In Admin → Manage Users
- Admin → Manage Spaces
- Admin → Manage Districts
- Admin → Approve Space Requests
- Admin → View All Bookings

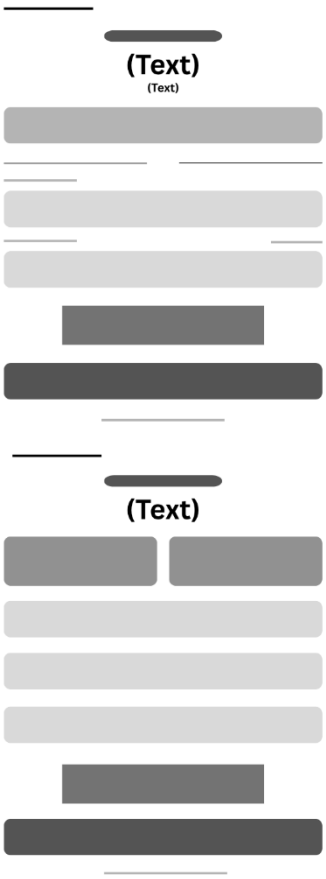
3. **Space owner**

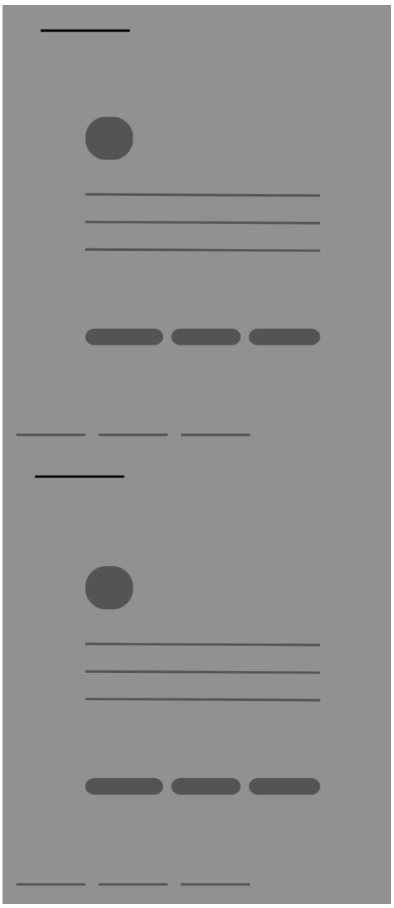
- Owner → Registers their co working-space
- Owner → Log In as space owner
- Owner →Manage their Space
- Owner → Manage Districts
- Owner → View bookings
- Owner → Approve bookings → Check in → Check out

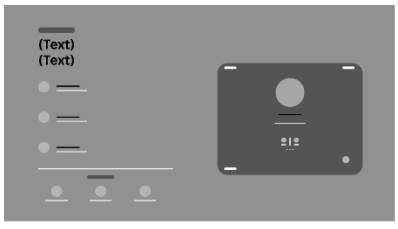
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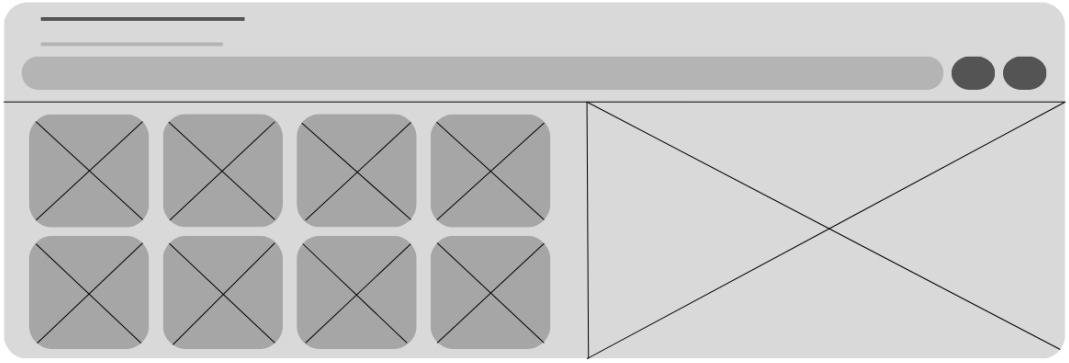


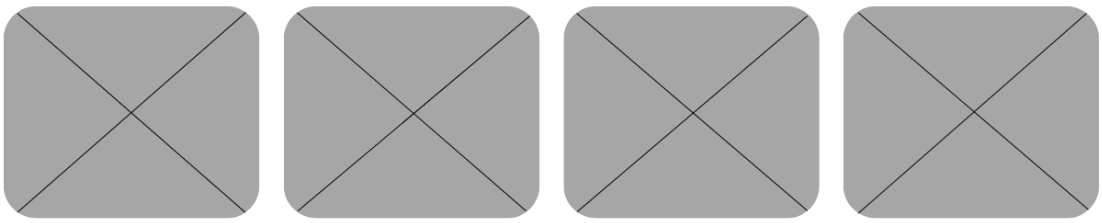

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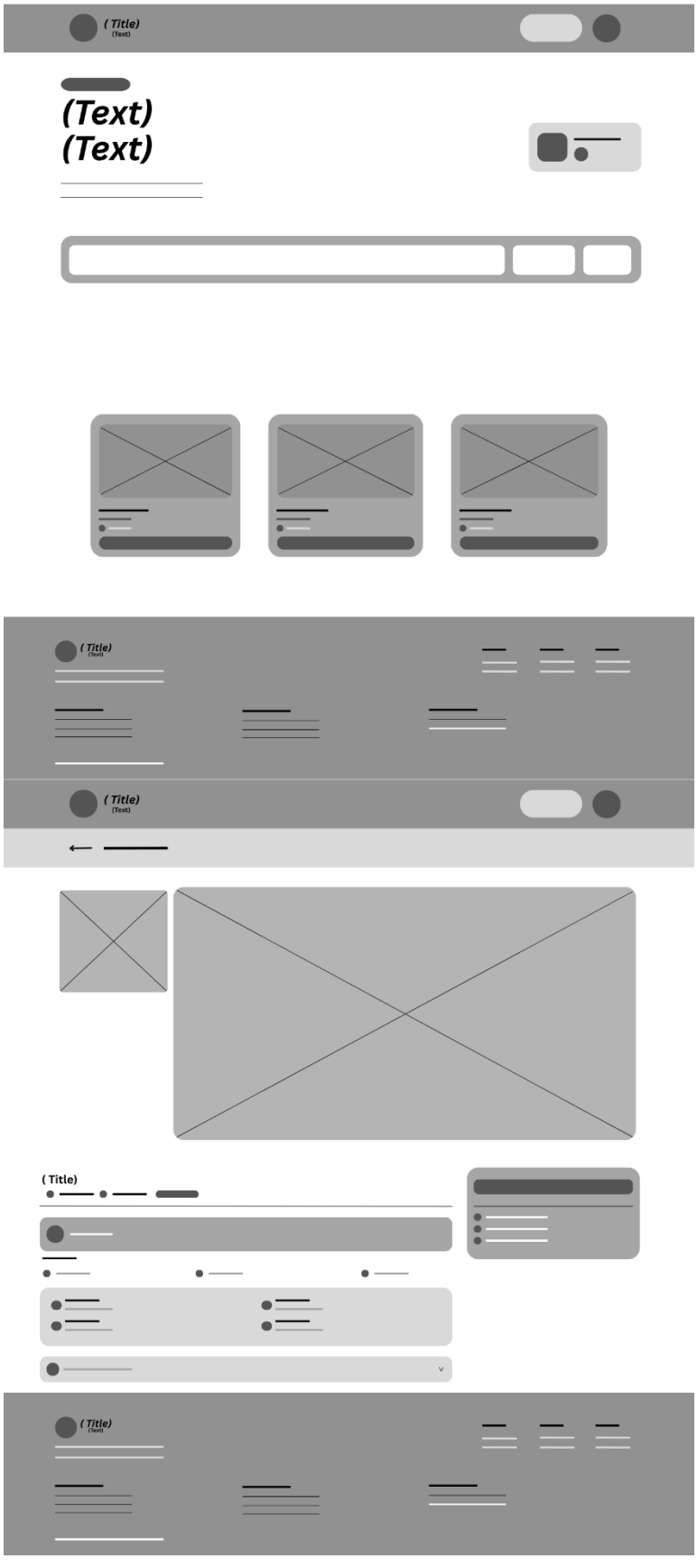
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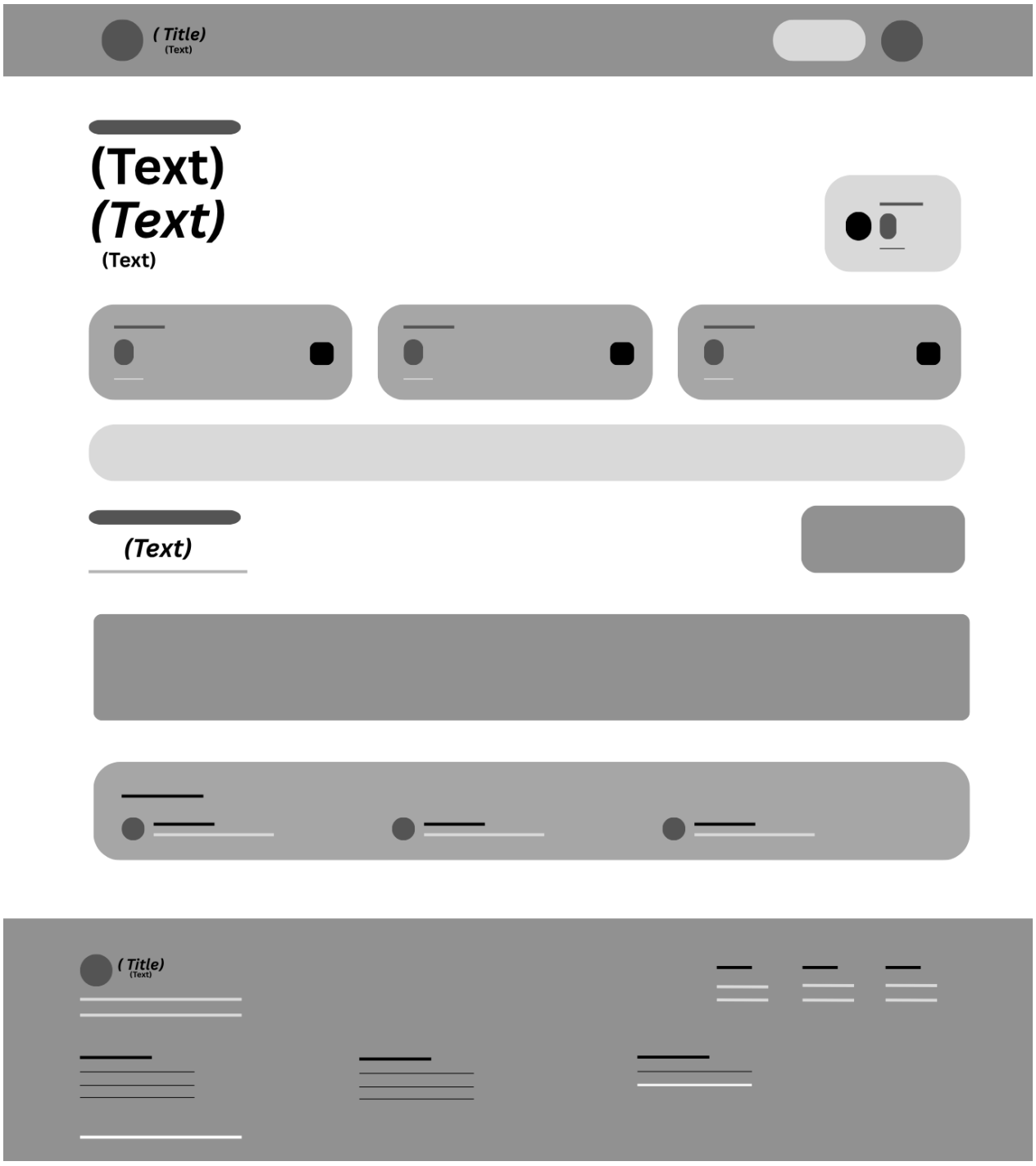


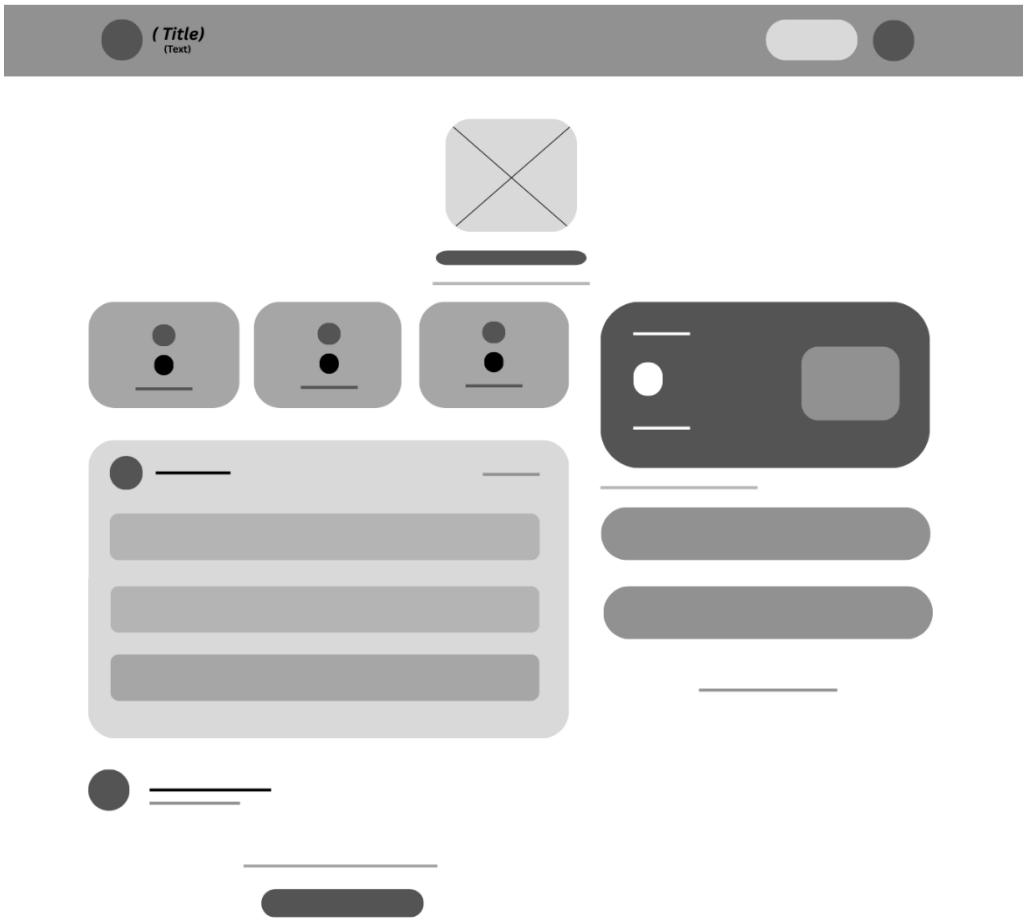


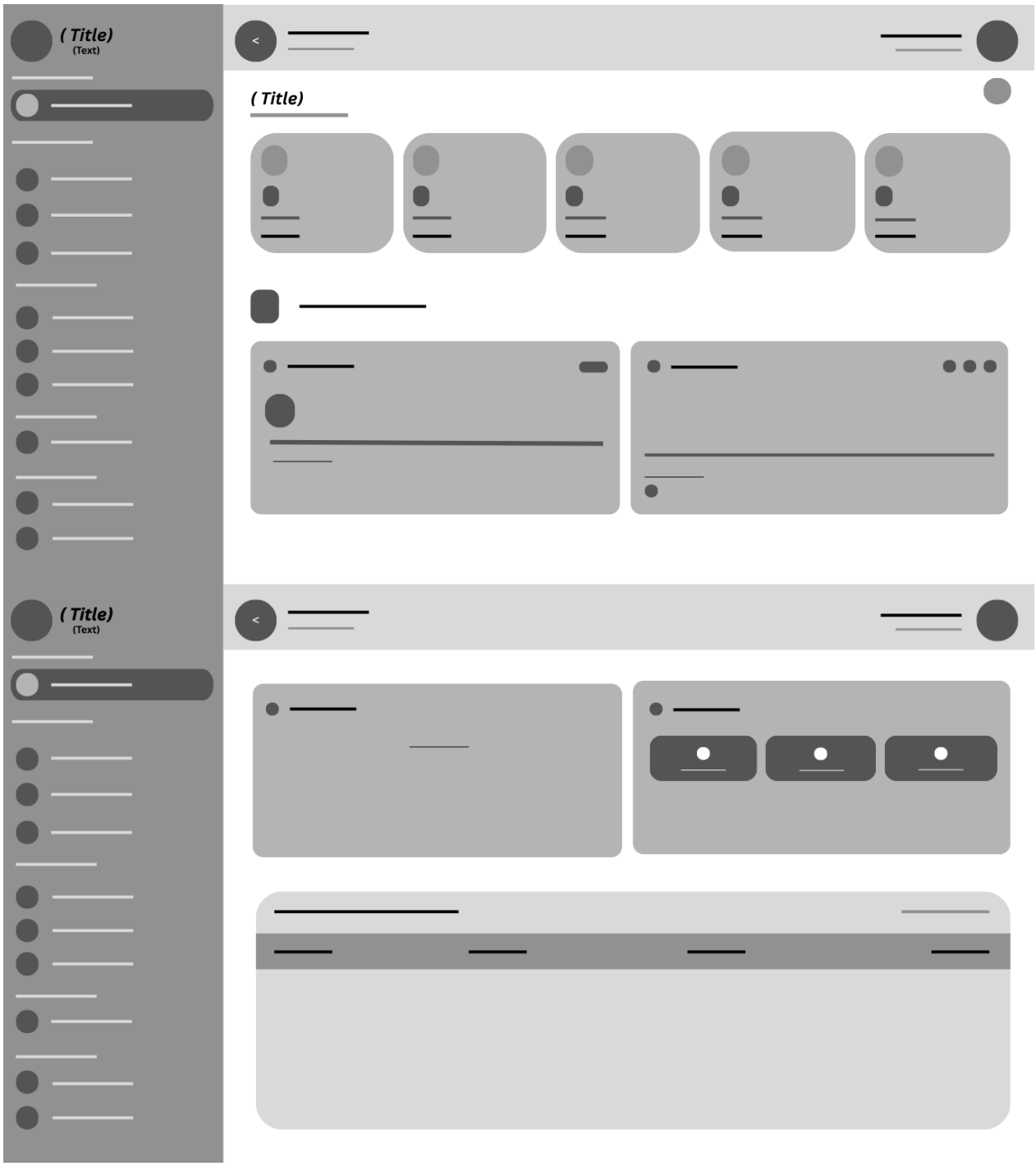
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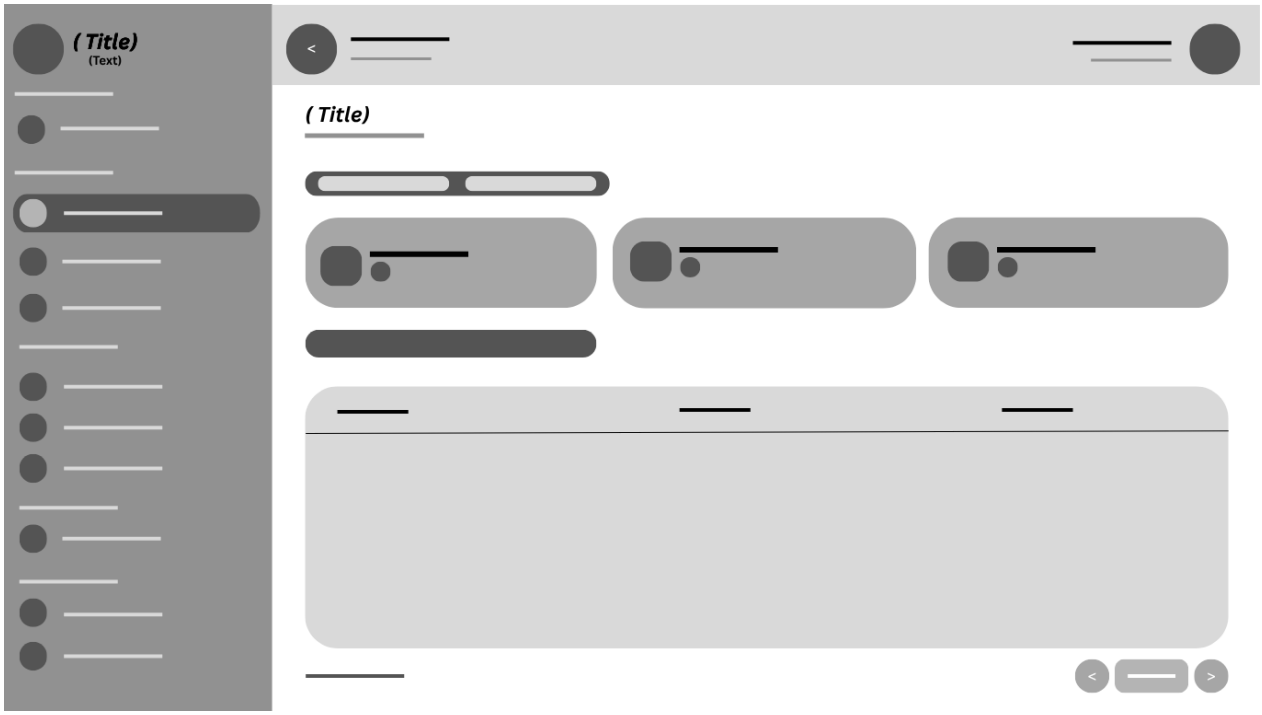

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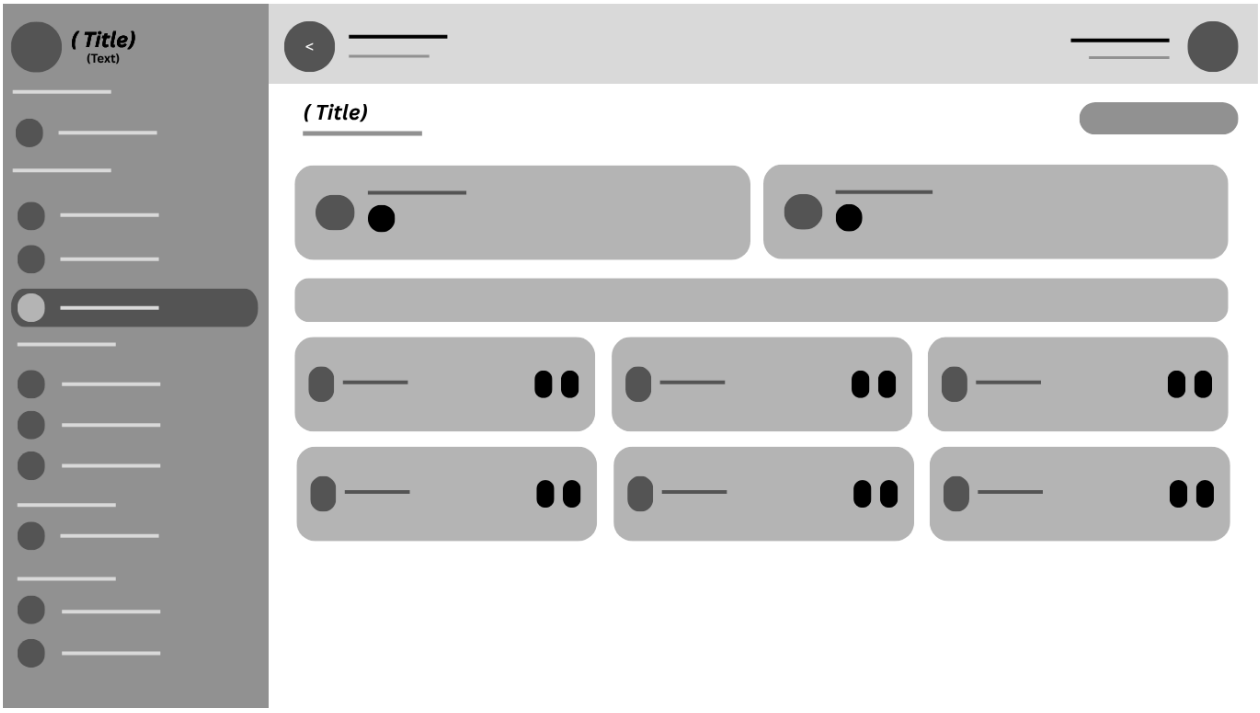
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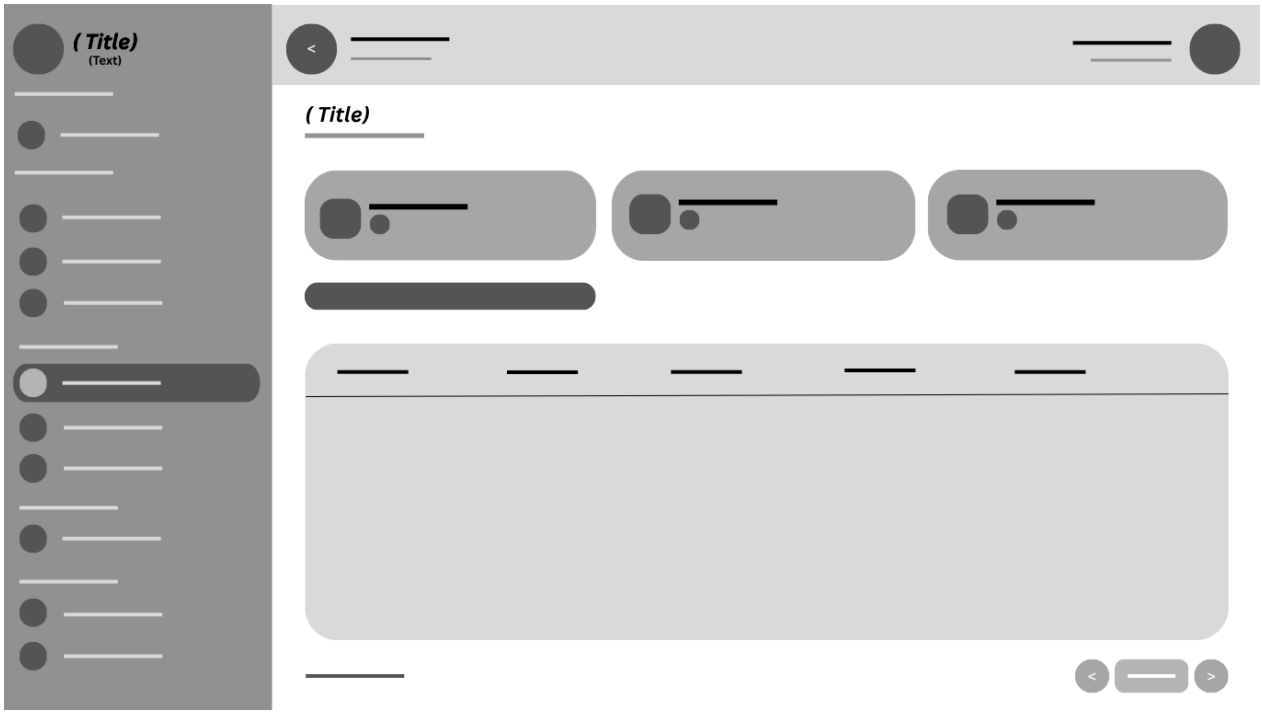

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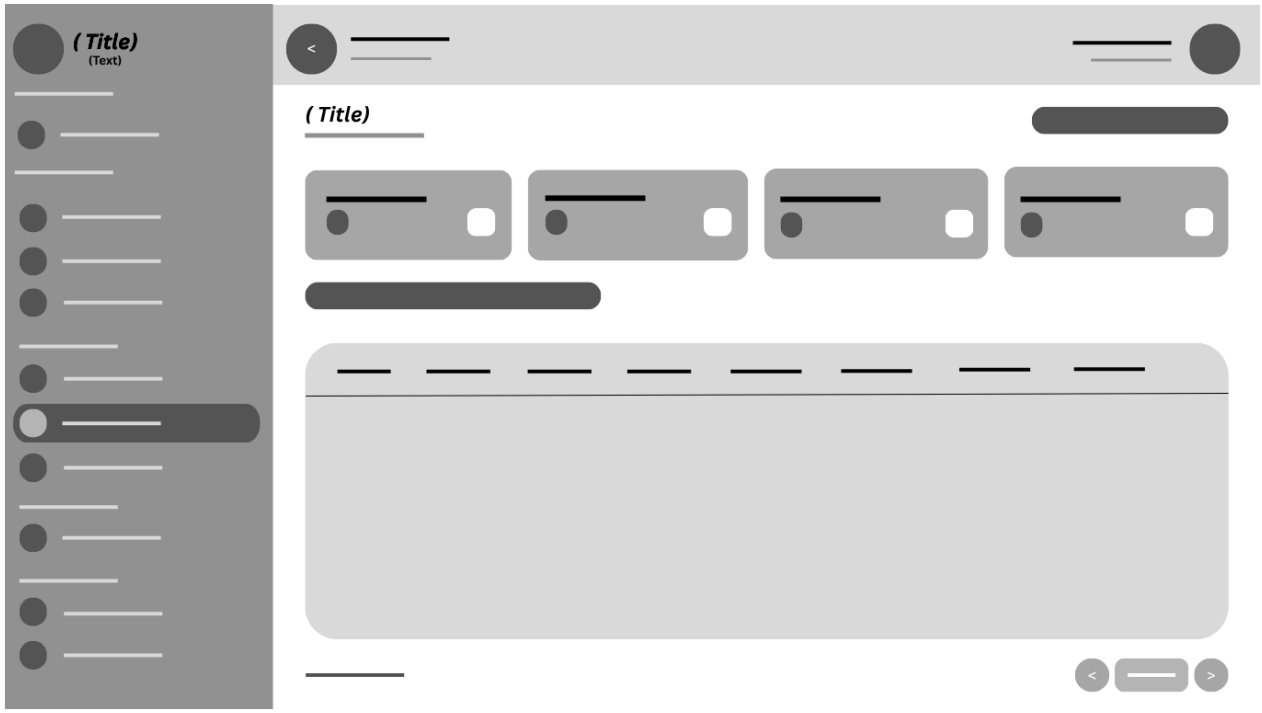

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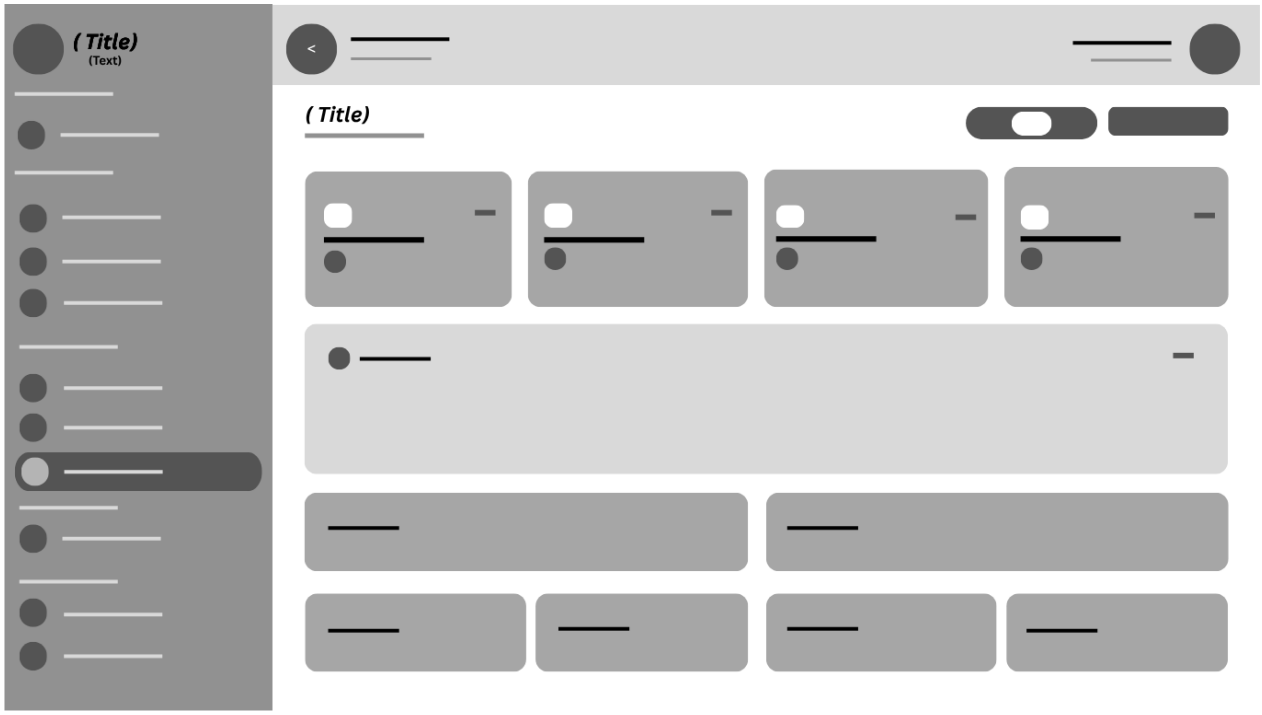

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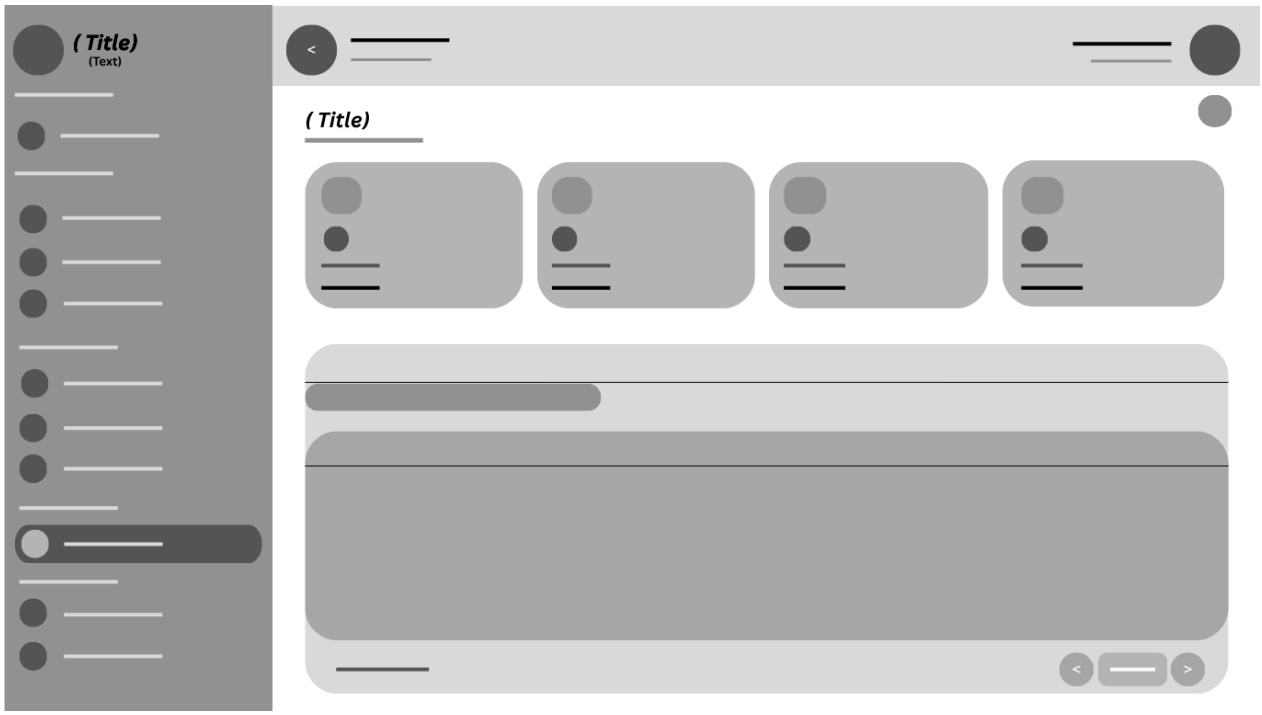

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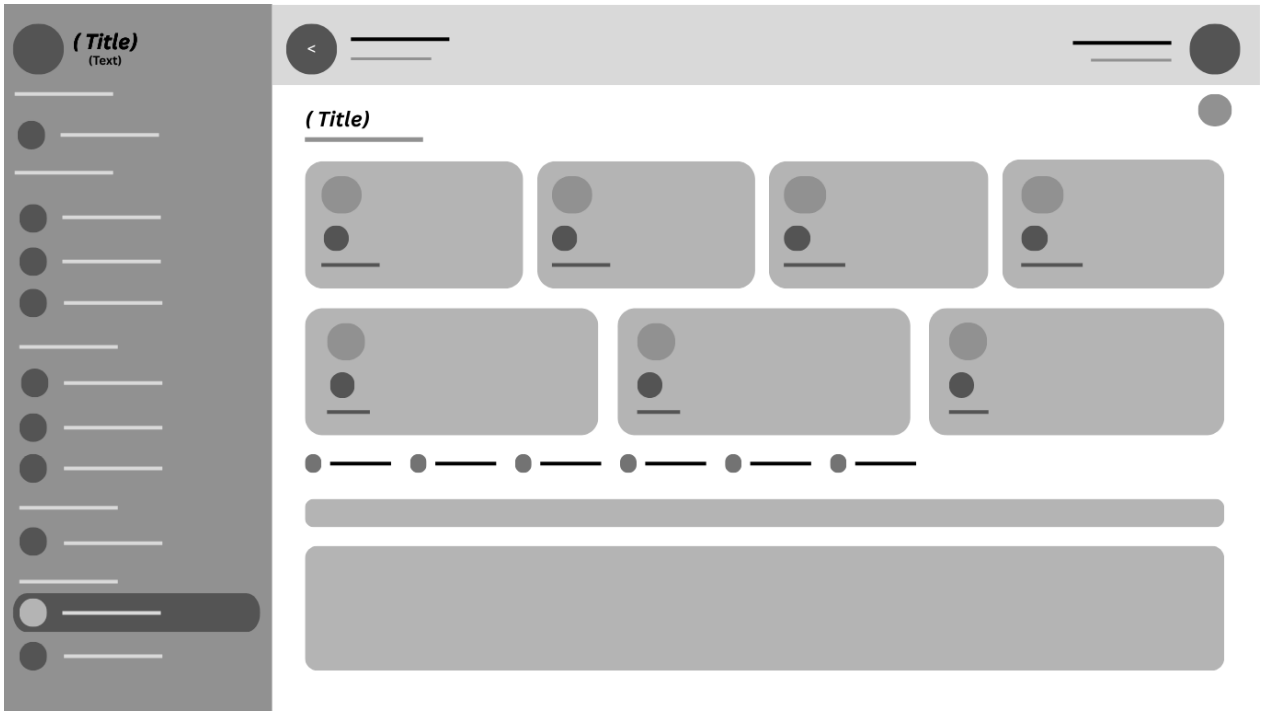

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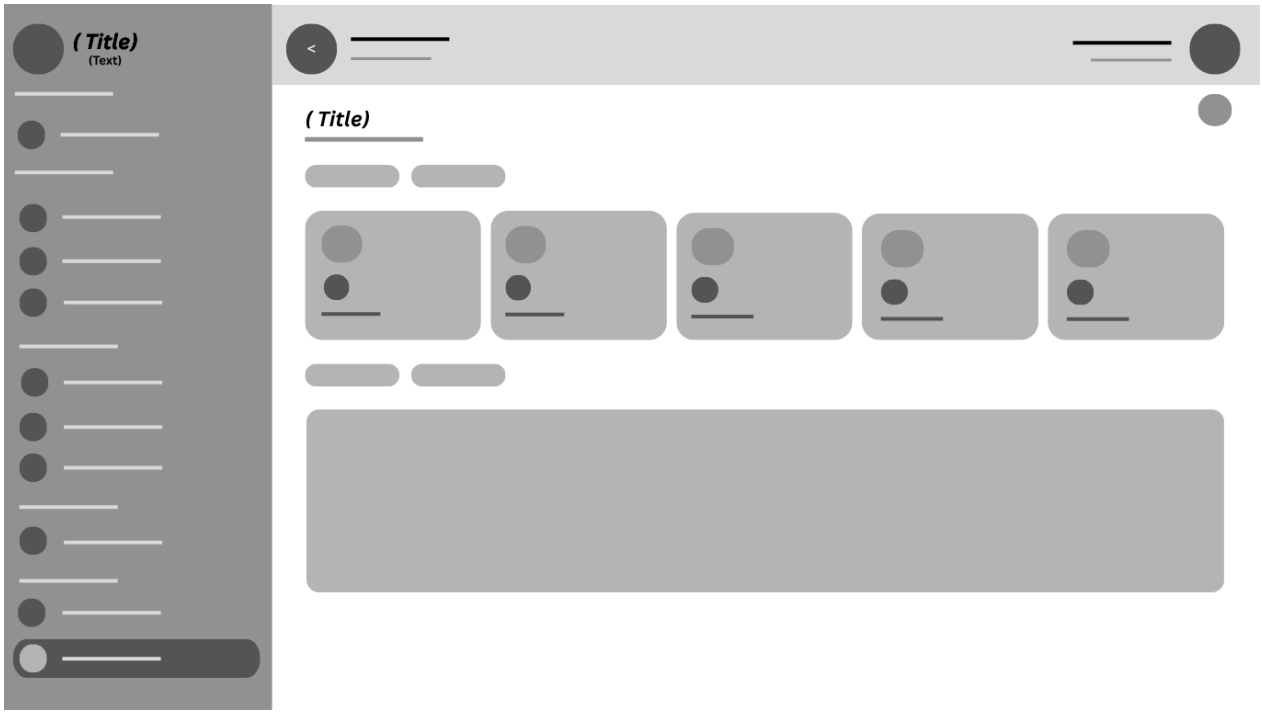

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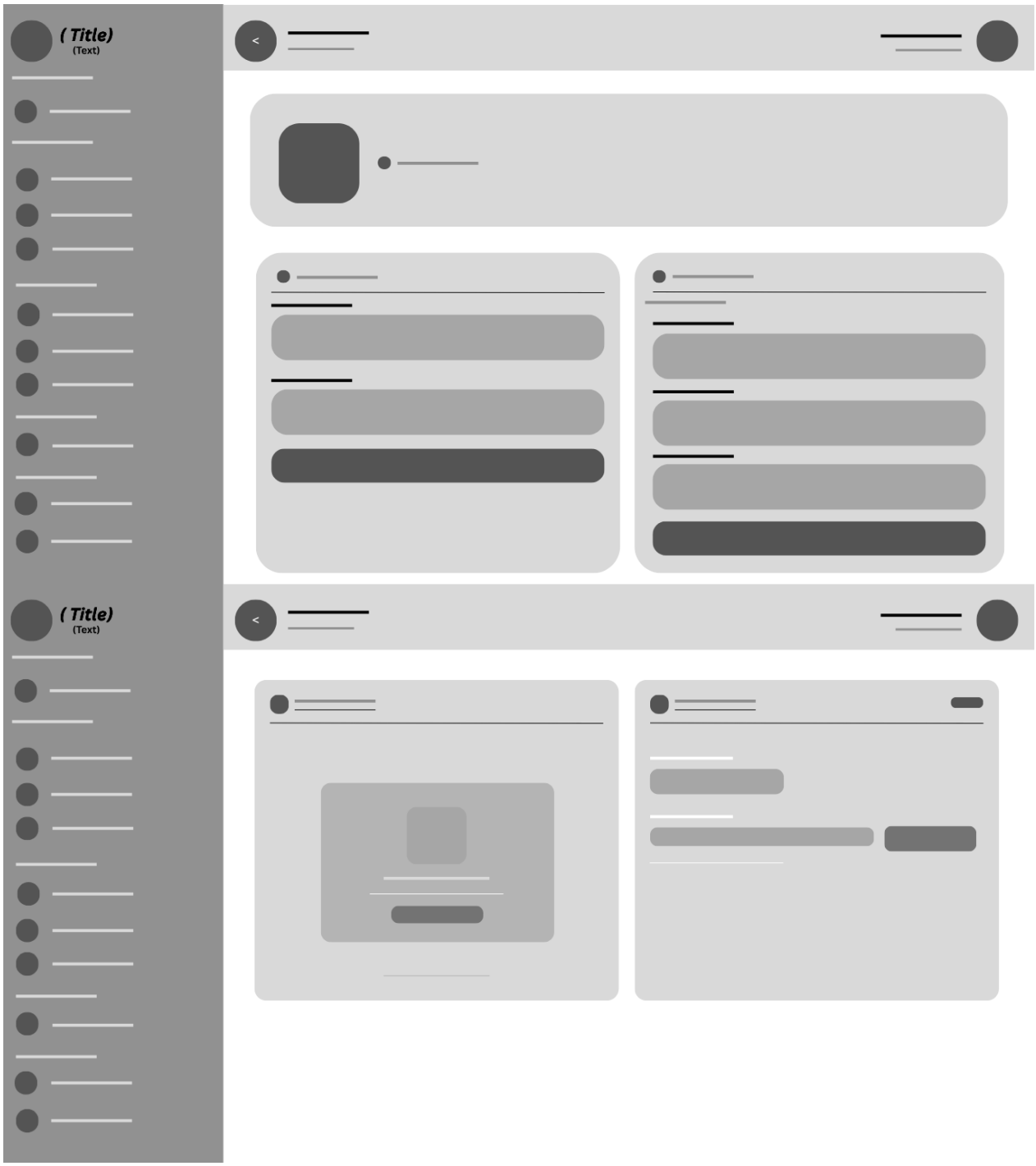

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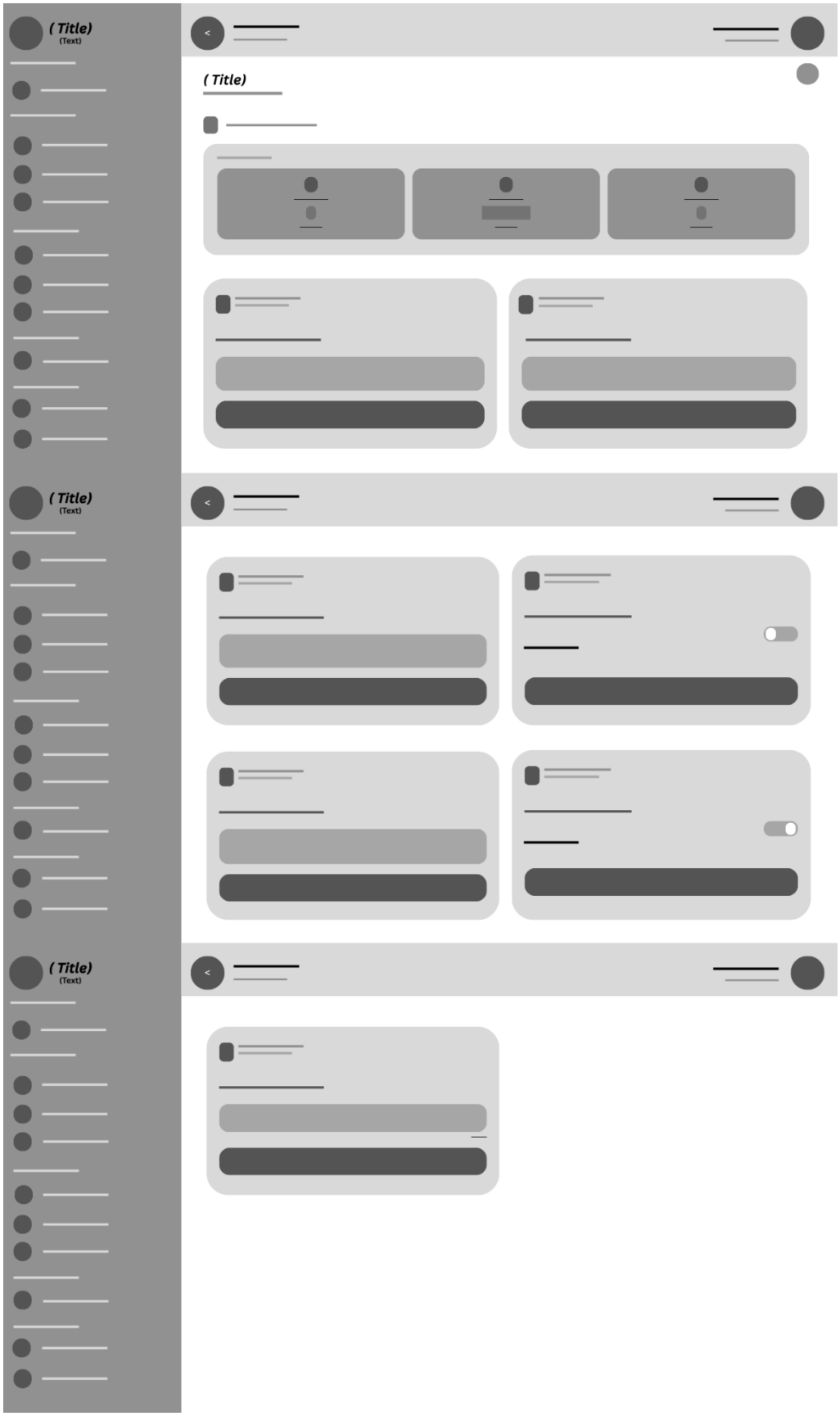

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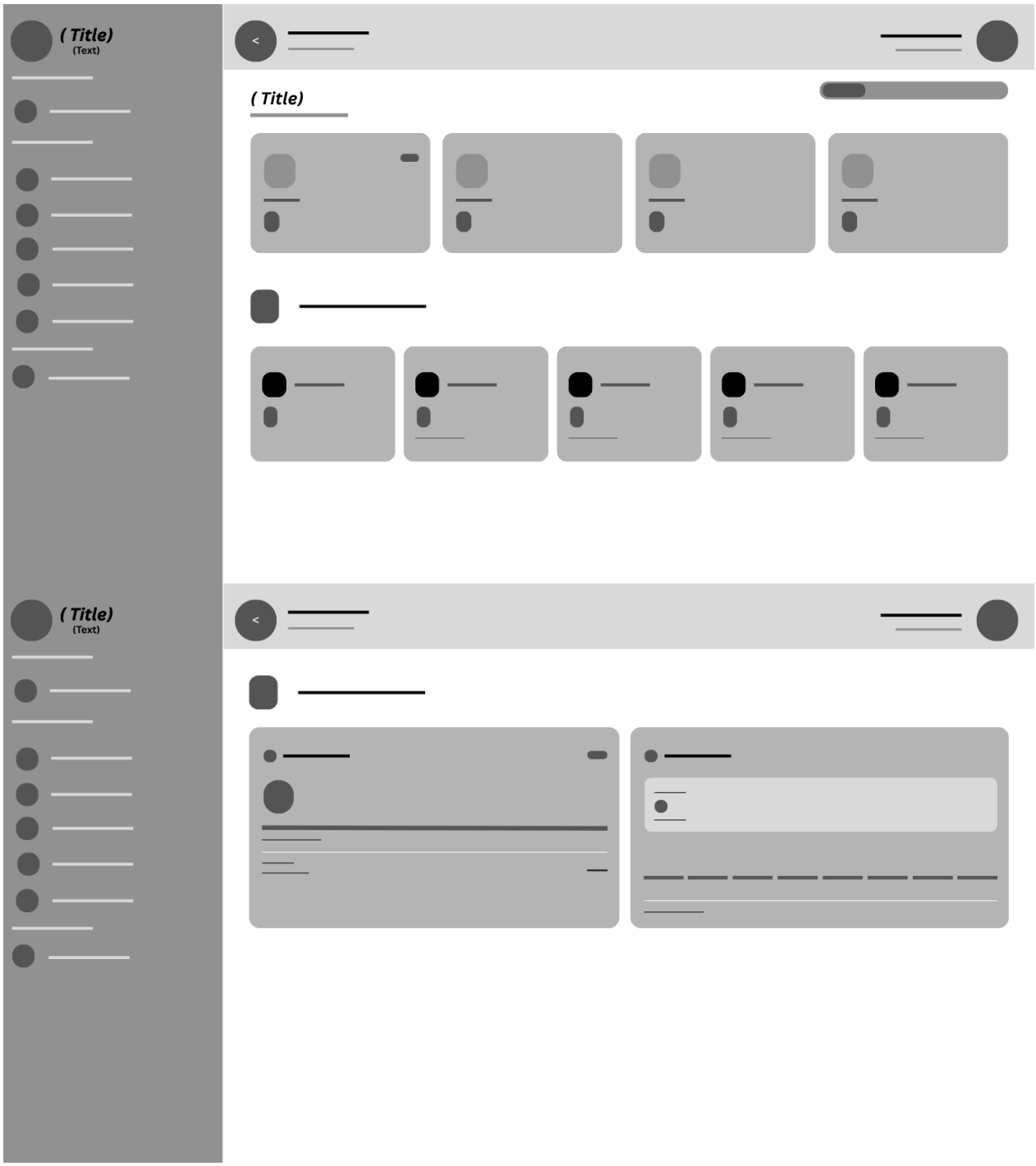

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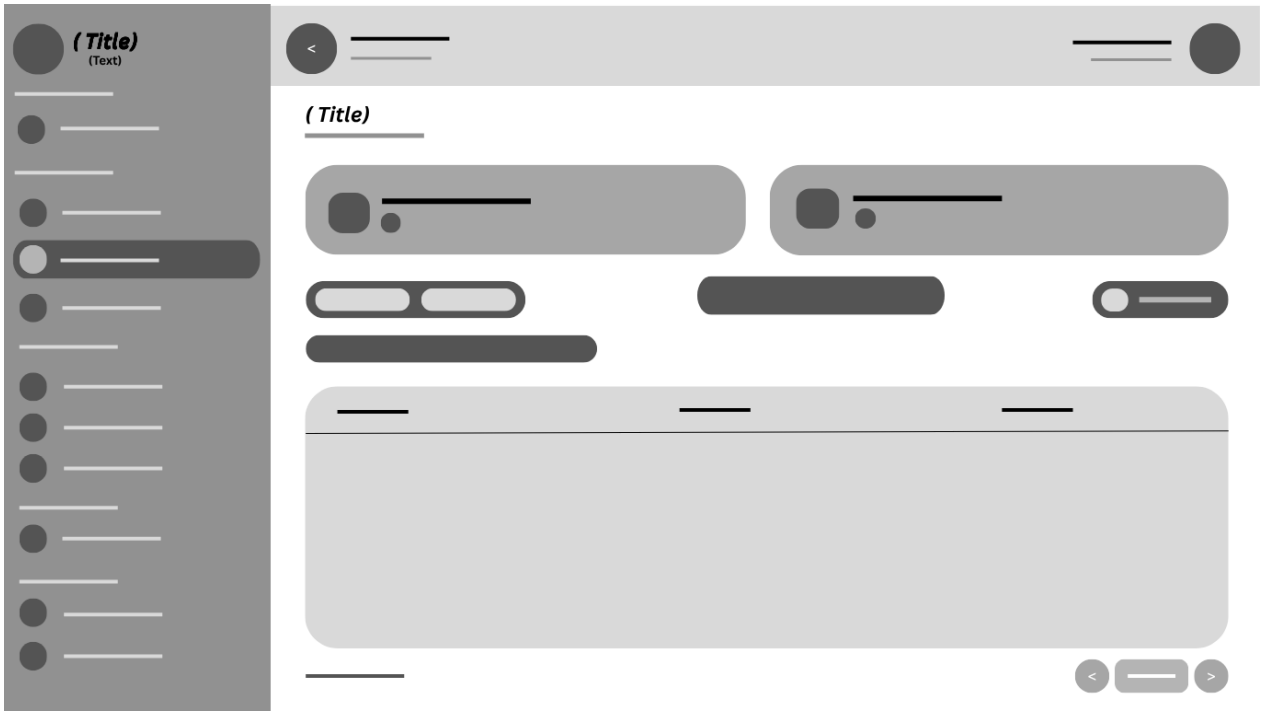

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**End of Proposal**

***Note to the instructors handling this subject***

**CHED BSIT ALIGNMENT STATEMENT**

This capstone project complies with CHED BSIT standards by demonstrating competencies in application development, database management, systems analysis and design, and emerging technologies such as machine learning. The project

follows outcomes-based education principles and integrates ethical considerations, user safety, and data privacy.